

Review

Natural language processing and computational linguistics for Vietnamese: A comprehensive review

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ABSTRACT

The success of modern Natural Language Processing (NLP), underpinned by large-scale Transformer architectures, is contingent upon access to extensive benchmark datasets. This dependence has intensified the resource disparity between a handful of high-resource languages and others, such as Vietnamese, where data scarcity poses a significant barrier to scientific and technological progress. To address this, we present the most extensive unified survey of Vietnamese NLP to date, distinguishing itself from prior task-specific overviews by simultaneously synthesizing 68 publicly available datasets, a complete architectural evolution from statistical to generative models, modern LLM benchmarks (VLUE, VMLU), and a structured linguistic challenge analysis within a single, consolidated reference. Each dataset is evaluated with respect to size, domain, annotation scheme, and accessibility. Beyond the dataset review, we synthesize the current state of Vietnamese NLP by examining 21 prominent modeling techniques from neural representations to Transformer-based architectures while benchmarking their empirical performance. Our analysis reveals critical gaps in available resources, particularly in under-represented domains and task categories, and identifies methodological challenges distinctive to Vietnamese, including word segmentation and syntactic ambiguity. In response, we propose a targeted roadmap with actionable recommendations for dataset creation, model innovation, and evaluation strategies. This survey aims to catalyze robust, reproducible research in Vietnamese, contributing to greater inclusivity within the global NLP community.

1. Introduction

Language is a structured system of symbols and rules that enables the communication of information. Natural Language Processing (NLP), situated at the intersection of Artificial Intelligence and Linguistics, empowers computers to comprehend and generate human language, thereby facilitating interaction with users without requiring expertise in programming languages. Given this intersection, it is crucial to distinguish NLP from its sibling discipline, Computational Linguistics (CL), as the two terms are often conflated despite their distinct goals. This fundamental difference, as articulated by Prof. Junichi Tsujii (Tsujii, 2021), is rooted in their distinct core objectives and parent disciplines. CL is positioned as a subfield of linguistics, a formal science concerned with creating a “formal or computational description of rules that languages follow”. Its primary goal is to model the abstract system of language with theoretical and mathematical rigor. In contrast, NLP is characterized as an engineering field focused on the practical application of “processing

of language by computer systems”. Consequently, the success of NLP is measured by the efficiency and effectiveness of working systems and is heavily constrained by available data and technology, whereas CL is driven by theoretical completeness and linguistic fidelity.

With this engineering focus in mind, NLP encompasses two principal domains: Natural Language Understanding (NLU) and Natural Language Generation (NLG). NLU addresses the interpretation of human language through areas such as phonology, morphology, syntax, semantics, and pragmatics, with foundational contributions from linguists like Noam Chomsky in syntactic theory. Conversely, NLG is concerned with producing coherent text from internal data structures, enabling machines to communicate information naturally.

NLP attracts interdisciplinary interest from computer science, linguistics, psychology, and philosophy, leading to the development of diverse theories and computational models for language interaction. Core NLP tasks include automatic summarization, coreference resolution, discourse analysis, machine translation, morphological segmentation,

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named entity recognition, optical character recognition, and part-of-speech tagging. These tasks are foundational to practical applications such as information extraction, conversational agents, and machine translation systems. While many of these tasks are interdependent sometimes functioning as components in larger processing pipelines each presents unique methodological challenges and research opportunities.

The landscape of NLP has been fundamentally transformed by the advent of Transformer architectures, first introduced by Vaswani et al. (2017). Originally developed for sequence-to-sequence machine translation, Transformers have rapidly become the dominant paradigm across a wide range of NLP tasks due to their effectiveness in modeling long-range dependencies and parallel processing capabilities. Landmark models such as BERT (Devlin et al., 2019) and GPT (Radford et al., 2018) have set new performance standards in contextual word representation, language modeling, and text generation, enabling significant advances in both understanding and generation tasks. BERT leverages a contrastive, masked language modeling objective, while GPT adopts a generative, autoregressive approach, together broadening the methodological toolkit for NLP research.

Recent developments have extended Transformer models to multimodal domains, integrating textual data with images and audio. These models excel at capturing cross-modal relationships, supporting tasks such as image captioning, text-to-image synthesis, and audio transcription. The versatility of Transformer-based approaches has led to their adoption in a variety of domains beyond traditional NLP.

Despite these advances, current NLP research remains heavily focused on high-resource languages, particularly English, often overlooking the diverse languages spoken by billions worldwide (Bender, 2019). Although deep learning and large-scale datasets have propelled progress in high-resource contexts, the scarcity of annotated data for low-resource languages like Vietnamese presents a persistent barrier (Ruder & Korashy, 2019). Addressing this challenge has become a central concern in contemporary NLP research.

The imperative to support low-resource languages has stimulated the development of resource-efficient techniques, including unsupervised learning, cross-lingual transfer, and data augmentation. These methods exploit alternative resources, such as unlabeled corpora or parallel data, and make varying assumptions about linguistic similarity and data availability. A clear understanding of these assumptions is crucial for selecting appropriate strategies for Vietnamese NLP.

Vietnamese, the national language of Vietnam, is spoken by nearly 100 million people, ranking as the 16th most common native language globally.¹ Despite this prevalence, Vietnamese is still considered a low-resource language in the context of natural language processing (NLP), especially when compared to high-resource languages such as English and Chinese. Over the past decade, significant progress in Vietnamese NLP has been achieved, largely through collaborative initiatives such as the four-year Vietnamese Language and Speech Processing (VLSP) project (2006–2010). In recent years, the availability of benchmark datasets for core tasks including word segmentation, part-of-speech tagging, dependency parsing (Nguyen et al., 2014a) and named entity recognition (NER) has substantially increased, particularly following the VLSP evaluation campaigns in 2013 and 2016. These efforts have stimulated a growing interest in building comprehensive NLP pipelines for Vietnamese, analogous to established toolkits like Stanford CoreNLP (Manning et al., 2014), thereby facilitating both foundational research and practical application development in this language.

This survey is intended for three primary audiences: (1) newcomers who require a clear and comprehensive introduction to Vietnamese NLP datasets; (2) researchers interested in constructing new datasets or advancing resource development for Vietnamese; and (3) NLP system developers seeking guidance on selecting, adapting, or transferring

datasets and models for specific architectures and tasks. The main contributions of this study are summarized as follows:

- **Dataset Taxonomy:** We consolidate 68 publicly available Vietnamese NLP datasets across 14 task categories into a unified structured taxonomy (Table 17) annotated with scale, domain, annotation method, primary evaluation metric, and accessibility status. Our gap analysis reveals that 6 of 14 task categories remain critically under-resourced (≤ 2 datasets), providing an evidence-based prioritization map for future corpus construction (Section 2).
- **Architectural Analysis:** We provide a systematic comparative analysis of Vietnamese NLP modeling paradigms from statistical CRFs through Bi-LSTM architectures and monolingual Transformers to regional large language models including a standardized performance comparison table (Table 18), explicit analysis of cross-lingual transfer strategies, and benchmarking on the VLUE and VMLU evaluation suites. The data-scaling hypothesis is confirmed empirically: ViDeBERTa's 138 GB pre-training corpus delivers consistent performance advantages across all five evaluated tasks (Section 3).
- **Research Roadmap:** Based on our structured gap analysis, we propose a concrete four-priority strategic roadmap specifying actionable directions for domain-specific Small Language Model (SLM) development via Parameter-Efficient Fine-Tuning (PEFT), multimodal integration using indigenous Vietnamese visual datasets, dialectal normalization pipelines, and hallucination mitigation through Retrieval-Augmented Generation each recommendation grounded in specific identified gaps (Section 4).

2. Vietnamese NLP background, task and dataset

Prior to delving into the specifics of this review, we provide a concise overview of the Vietnamese natural language processing (NLP) undertaking. We categorize the current NLP task as follows. There are two basic categories for all datasets in this survey: Core task and Application task. We split datasets into two groups based on their core and application tasks.

The empirical maturation of Vietnamese NLP is inextricably tethered to the curation of high-fidelity annotated corpora. Over the past decade, the field evolved from a severe scarcity of basic resources to a rich ecosystem of specialized benchmarks, driven primarily by the annual VLSP evaluation campaigns (2013–2025). Table 17 consolidates all publicly available datasets into a unified taxonomy, revealing persistent structural gaps: Relation Extraction (1 dataset), Semantic Similarity (2 datasets), and Word Segmentation (2 datasets) remain critically under-resourced relative to their importance as foundational pipeline components. Fig. 1 visualizes this imbalance explicitly.

2.1. Core NLP tasks

2.1.1. Part-of-speech tagging

Part-of-speech tagging (POS tagging) in Vietnamese involves labeling each word in a sentence with its corresponding part of speech, such as noun, verb, adjective, etc. This is important for various natural language processing tasks, such as machine translation and text-to-speech synthesis. Vietnamese is a morphologically rich language, which poses unique challenges for part-of-speech tagging due to a large number of possible word forms and the lack of explicit markers for some parts of speech. Let W be the words $w_1, w_2, \dots, w_n$ in a sentence and T be their corresponding POS tags $t_1, t_2, \dots, t_n$. The goal of POS tagging is to predict the POS tags $t_1, t_2, \dots, t_n$ for the words $w_1, w_2, \dots, w_n$. The general formula for POS tagging can be expressed as follows:

$$T = \operatorname{argmax}_T P(T|W) \quad (1)$$

where $P(T|W)$ is the conditional probability of the POS tag sequence given the word sequence. The formula can be interpreted as selecting the most likely sequence of POS tags given the observed words in the

¹ www.worldometers.info/world-population/vietnam-population/

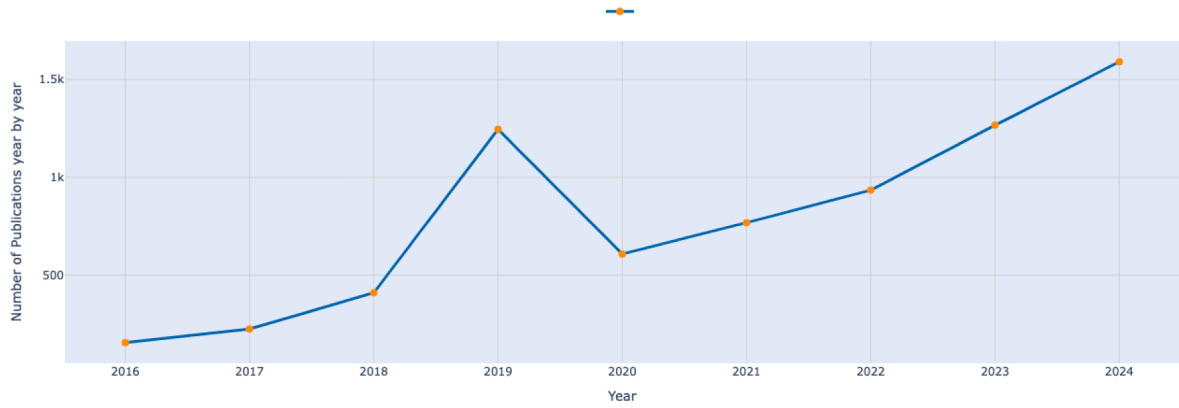


Figure 1. The trend of research publications in the field of Vietnamese Natural Language Processing (NLP) over a span of ten years, from 2016 to 2024. The data for this analysis was obtained from app.dimensions.ai. The steep growth between 2021 and 2024 reflects the expanding scope of Vietnamese NLP beyond core linguistic tasks toward application domains such as legal engineering, fact-checking, and multimodal question answering.

Table 1
Comparison of modeling paradigms for Vietnamese POS tagging.

Paradigm	Advantages	Limitations
HMM / MaxEnt	Low data requirement; fast inference; interpretable	Hand-crafted features; brittle to OOV and domain shift
BiLSTM-CRF	Contextual features; char-level OOV handling; no manual feature engineering	Needs moderate labeled data; fixed context window
PhoBERT finetune	SOTA 97.2%; transfer learning from large Vi corpus; flexible task adaptation	RDRSegmenter preprocessing dependency; high GPU memory; degrades on informal text
LLM prompting	Zero-shot capable; no task-specific annotation required	Inconsistent on structured label output; no canonical Vietnamese POS evaluation

sentence. The probability $P(T|W)$ is typically computed using a statistical model such as a Hidden Markov Model (HMM) or a neural network-based model.

Methodological Evolution

Research on Vietnamese POS tagging has advanced through three distinct paradigm shifts. Early systems employed Hidden Markov Models (HMMs) and Maximum Entropy classifiers, which required manually engineered lexical and contextual feature templates but achieved reasonable accuracy on the limited corpora available at the time. The introduction of BiLSTM-CRF architectures substantially improved contextual disambiguation by jointly modelling tag-sequence dependencies alongside character-level representations, reducing the reliance on hand-crafted features while better handling Vietnamese out-of-vocabulary tokens. Most recently, PhoBERT-based finetuning has established a performance ceiling of 97.2% accuracy on the VLSP 2013 benchmark (Nguyen & Tuan Nguyen, 2020), confirming that large-scale Vietnamese pretraining effectively subsumes the feature engineering burden of prior approaches. Table 1 summarises the trade-offs across these paradigms.

The advancement of Vietnamese POS tagging has been closely tied to the development of annotated corpora. A pivotal resource in this regard is the VLSP2013 POS Tagged Corpus, which consists of approximately 30,000 sentences sourced from Vietnamese online news. Each sentence in this corpus is meticulously tokenized and annotated with POS tags, providing a robust foundation for both supervised learning and benchmarking.

Recognizing the necessity for richer linguistic resources, subsequent efforts culminated in the creation of the NIIVTB (National Institute of Information and Communications Technology Vietnamese Treebank) (Nguyen et al., 2018d). NIIVTB extends beyond basic POS annotation by incorporating detailed word segmentation and syntactic bracketing, thus supporting comprehensive syntactic analysis. The corpus comprises 20,588 sentences, divided into two subsets that reflect diverse topical domains one from Tuoitre news, emphasizing socio-political content,

and another from Thanhnien news, covering 14 distinct topics with balanced representation.

In summary, the VLSP2013 and NIIVTB corpora exemplify the trajectory of Vietnamese NLP resources, evolving from essential POS annotation toward more sophisticated, multi-layered syntactic representations. This progression not only underpins the development of high-performance POS taggers but also facilitates broader advancements in Vietnamese computational linguistics.

2.1.2. Named entity recognition

Named Entity Recognition (NER) is a natural language processing task that aims to identify and classify named entities in text into predefined categories such as person names, organization names, and locations. NER has been applied to various languages, including Vietnamese. NER in Vietnamese is a challenging task due to the complex structure of Vietnamese language, including the use of multiple accents and diacritics, as well as the lack of clear delimiters between words. However, recent research has shown promising results in NER for Vietnamese using advanced machine learning techniques. Given an input sequence of words $w_1, w_2, \dots, w_n$, the goal is to assign a label y_i to each word w_i that corresponds to an entity type. This can be represented by the following formula

$$(w_1, w_2, \dots, w_n) \rightarrow (y_1, y_2, \dots, y_n) \quad (2)$$

where y_i is the entity label assigned to the word w_i .

Methodological Evolution

Vietnamese Named Entity Recognition has followed a clear evolutionary trajectory from rule-based heuristics to state-of-the-art Transformer models. Early approaches combined handcrafted gazetteer features with Conditional Random Fields (CRF), exploiting Vietnamese capitalization conventions and known entity lists to achieve moderate precision. BiLSTM-CRF models subsequently replaced manual features with automatically learned character and word embeddings, improving recall on multi-word entity spans and out-of-vocabulary named en-

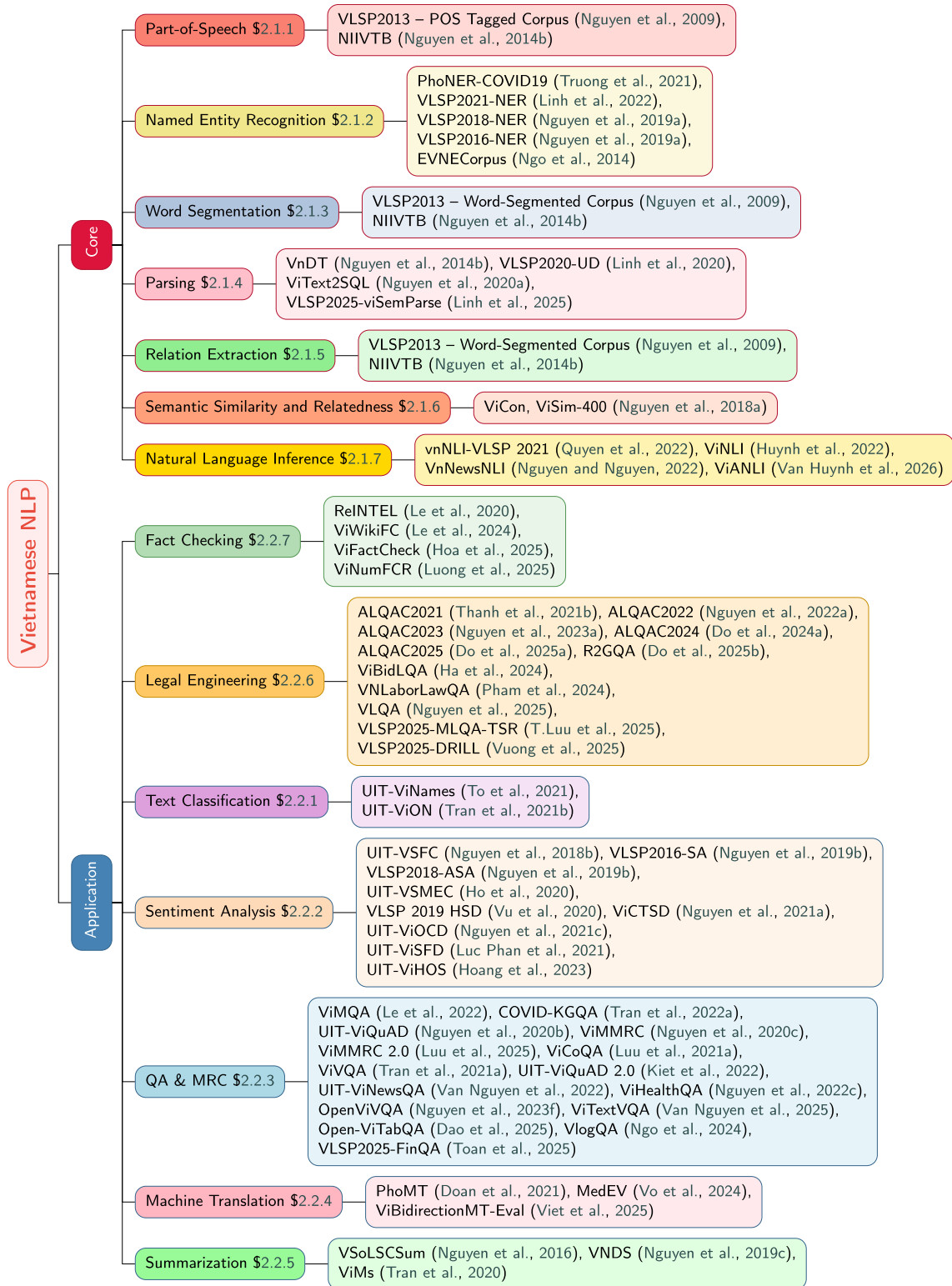


Figure 2. An illustration of typical tasks and datasets in Vietnamese NLP in 14 task categories spanning core linguistic processing and applied NLP.

tities. The release of PhoBERT marked a decisive shift: by encoding long-range contextual dependencies through self-attention, PhoBERT-based systems substantially outperformed recurrent models on all VLSP NER benchmarks. The most recent best result of 95.3 F1 is achieved by ViDeBERTa, whose 138 GB pretraining corpus provides richer Vietnamese entity representations than earlier models. Table 2 compares

these paradigms, and Table 3 reports benchmark scores across representative systems.

Table 3 demonstrates a consistent hierarchy in which Vietnamese-specific pretraining outperforms multilingual models at every scale: PhoBERT-base surpasses XLM-R-large on VLSP 2018 despite having fewer parameters, confirming that dedicated Vietnamese pretraining

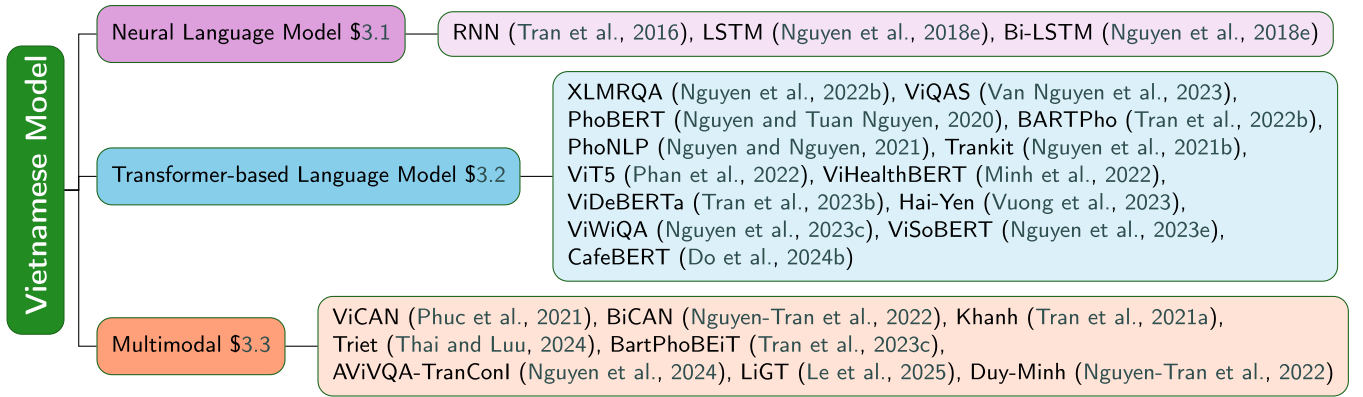


Figure 3. An illustration of typical tasks and datasets in Vietnamese.

Table 2
Comparison of modeling paradigms for Vietnamese NER.

Paradigm	Advantages	Limitations
CRF + gazetteer	High precision on known entities; no large training set required	Low recall on novel/nested entities; manual feature engineering required
BiLSTM-CRF	Automatic feature learning; handles multi-word spans; char-level OOV robustness	Fixed context window; requires moderate annotated data; sensitive to tokenization errors
PhoBERT finetune	SOTA on VLSP benchmarks; rich Vietnamese contextual representations; strong transfer across entity types	RDRSegmenter dependency; high compute; degrades on diacritic-dropped text
ViDeBERTa finetune	Current SOTA 95.3 F1; benefits from 138 GB pretraining; best cross-task generalization	Largest memory footprint; pretraining corpus details not fully documented

Table 3
Model performance on Vietnamese NER benchmarks (F1 score, %). Best results are **bolded**.

Model	VLSP 2016	VLSP 2018	VLSP 2021	PhoNER-COVID19
CRF baseline	88.2	74.4	–	–
BiLSTM-CRF	91.3	76.6	–	–
mBERT	91.5	79.1	–	82.3
XML-R-large	93.6	84.0	–	87.1
PhoBERT-base	94.7	87.7	80.1	91.4
PhoBERT-large	95.1	88.6	81.3	92.7
ViDeBERTa	95.3	89.1	82.9	93.8

provides a stronger representational prior than multilingual capacity alone. ViDeBERTa's margin over PhoBERT-large further supports the data-scaling hypothesis: the additional 118 GB of pretraining text (138 GB vs. 20 GB) yields consistent improvements across all four benchmarks.

The **story of Vietnamese NER** begins with the **EVNECorpus** (Ngo et al., 2014), constructed using a **semi-automatic** approach and comprising over 45,000 sentences. Within this corpus, there are 32,454 English named entities and 33,338 Vietnamese named entities, classified into six categories: Location (LOC), Person (PER), Organization (ORG), Time (including date tags) (TIM), Money (MON), and Percentage (PCT). The annotation process starts with English texts, which are tagged using the Stanford NER system. These annotated English texts are then aligned with their Vietnamese counterparts, after which the Vietnamese entity tags are meticulously revised and corrected to ensure high accuracy and consistency.

Subsequent progress in the domain was marked by the introduction of the **VLSP2016-NER** corpus (Nguyen et al., 2019a), which provided 16,000 sentences annotated for three core entity types: persons (PER), organizations (ORG), and locations (LOC). Each sentence was not only manually segmented into words but also enriched with automatically generated part-of-speech (POS) and phrase chunking labels. The data,

sourced from electronic news articles, reflects contemporary Vietnamese usage and supports both sequence labeling and syntactic analysis tasks.

The trajectory of Vietnamese NER was further propelled by the release of **VLSP2018-NER** (Nguyen et al., 2019a) and **VLSP2021-NER** (Linh et al., 2022). While the 2016 and 2018 datasets focused on three principal entity categories, the 2021 edition introduced a substantially **broader challenge**, encompassing **14 main entity types** and **26 sub-types**. The VLSP2021-NER resource is distributed in two packages: one containing the complete 2018 dataset and another with newly collected articles from major Vietnamese news platforms such as **vnexpress.vn**, **baomoi.com**, and **zingnews.vn**. This expansion addressed previous gaps by including additional entity types and diverse domains.

In direct response to the COVID-19 pandemic, the **PhoNER-COVID19** dataset (Truong et al., 2021) was developed as a specialized resource, containing over **35,000 annotated entities across 10,000 sentences**. It defines **10 distinct entity types relevant to COVID-19**, enabling precise extraction of critical information for healthcare and epidemiological applications.

These corpora together illustrate the evolution of Vietnamese NER resources, demonstrating the research community's sustained commitment to expanding linguistic coverage, annotation depth, and practical utility. Each dataset incrementally enhances annotation richness and domain diversity, thereby supporting both foundational research and the development of advanced NER systems for Vietnamese. Current best performance is 95.3% F1 by **ViDeBERTa** (Tran et al., 2023b), confirming the benefit of large-scale pre-training even on established benchmarks.

2.1.3. Word segmentation

Word segmentation in Vietnamese is a crucial task for natural language processing as **Vietnamese is a non-space language, meaning that words are not separated by spaces as in English**. Vietnamese text must be segmented into words before any further analysis can be done. Given an input sentence S consisting of a sequence of characters $S = (c_1, c_2, \dots, c_n)$, the task is to segment the sentence into a sequence of words $W = (w_1, w_2, \dots, w_m)$, where each word w_i is a sequence of characters $w_i = (c_{i1}, c_{i2}, \dots, c_{ik})$, with $1 \leq i \leq m$ and $\sum_{i=1}^m k_i = n$. The task is to

Table 4
Comparison of Vietnamese word segmentation approaches.

Approach	Advantages	Limitations	Tools / Work
Dictionary / max-match	Very fast; no training data required; deterministic output	OOV failure; no contextual disambiguation; compound noun errors	Early VLSP tools
CRF sequence labelling	Contextual; handles ambiguous spans; data-driven feature learning	Requires annotated corpus; fixed feature templates; brittle to unseen patterns	VLSP 2013 system
Neural / rule-list hybrid	High accuracy (>97% F1); practical inference speed	News-domain only; degrades on informal/social media text; tools produce non-equivalent outputs	RDRSegmenter, Vn-CoreNLP, pyvi
Syllable-level Trans-former	No segmentation preprocessing required; end-to-end training; robust to informal text	Longer input sequences; higher compute cost per token	BARTpho-syllable

finding the optimal word sequence W^* that maximizes the conditional probability $P(W|S)$, which can be computed using Bayes' rule as:

$$P(W|S) = \frac{P(S|W)P(W)}{P(S)}, \quad (3)$$

where $P(S|W)$ is the probability of the input sequence given the word sequence, $P(W)$ is the prior probability of the word sequence, and $P(S)$ is the probability of the input sequence.

Methodological Evolution

Word segmentation is a prerequisite processing step for virtually all downstream Vietnamese NLP tasks, making it one of the most foundational yet persistently challenging components of the Vietnamese NLP pipeline. Early approaches relied on dictionary-based maximum matching algorithms, which greedily tokenized input syllable sequences by matching against known lexicons. While computationally efficient, these systems failed on out-of-vocabulary compound nouns and named entities not present in the dictionary. Statistical CRF-based segmenters such as the VLSP 2013 system improved robustness by framing segmentation as a sequence labelling problem over syllable characters, incorporating contextual syllable n-gram features. Contemporary tools including RDRSegmenter, VnCoreNLP, and pyvi adopt neural or rule-list approaches that achieve over 97% F1 on the VLSP 2013 benchmark. However, a critical gap persists: all major segmenters are trained exclusively on formal news-domain text, and their performance degrades substantially on social media, where diacritic dropping and code-switching produce syllable sequences not observed during training. Table 4 summarises the trade-offs among current approaches.

VLSP2013 – Word-Segmented Corpus (Nguyen et al., 2009) is a first accessible dataset consisting of 75,000 training sentences that were manually segmented into words as part of the VLSP 2013 word segmentation shared task. Additionally, the dataset also includes 2120 test sentences from the VLSP 2013 POS tagging shared task. The VLSP2013 Word-Segmented Corpus remains the primary benchmark; however, with only 2 publicly available datasets, this task is critically under-resourced relative to its pipeline importance, and the sole reliance on a 2013 corpus raises reproducibility concerns for evaluating models on contemporary internet and social media language.

2.1.4. Syntax parsing

Syntax Parsing in Natural Language Processing (NLP) refers to the process of analyzing the grammatical structure of a sentence to determine its syntactic organization according to a given formal grammar. The primary objective of parsing is to map a linear sequence of words into a hierarchical structure, often represented as a parse tree, which reveals the relationships among words and constituents within the sentence. In Vietnamese Natural Language Processing, parsing is generally categorized into two main types: dependency parsing and semantic parsing.

Dependency parsing in Vietnamese is the task of analyzing the grammatical structure of a sentence to identify the dependencies between words. The task of dependency parsing is to find the optimal set of dependency relations that form a valid dependency tree for the input

sentence. This can be represented by the following math formula:

$$T = \operatorname{argmax}_{T' \in \mathcal{T}} \prod_{(i,j,d) \in T'} p(i,j,d|S) \quad (4)$$

where T is the set of all valid dependency trees for the input sentence, $\mathcal{T}$ is the space of possible trees, and $p(i,j,d|S)$ is the probability of a specific dependency relation (i,j,d) given the sentence S (Figs. 2, 3).

Methodological Evolution

Vietnamese dependency parsing has evolved from transition-based parsers with hand-crafted features to neural graph-based approaches with pretrained contextual representations. Early parsers applied the MaltParser transition system with Vietnamese-specific morphosyntactic features, exploiting the VnDT treebank as the primary training resource. Graph-based neural parsers using BiLSTM encoders subsequently improved attachment scores by modelling global tree structure rather than local transition sequences, reducing error propagation in long sentences. The adoption of PhoBERT as a contextualized encoder for Vietnamese parsing further improved performance, as the pretrained model provides implicit knowledge of Vietnamese argument structure that reduces dependence on explicit feature engineering (Tables 21, 23). A key unresolved challenge is the scarcity of Vietnamese treebank data: the VLSP 2020-UD corpus contains approximately 9000 sentences, which is roughly one-tenth the size of the Penn English Treebank, limiting parser accuracy on complex and rare syntactic constructions. Table 5 summarises the methodological progression.

One of the foundational contributions is the **VnDT** (Nguyen et al., 2014b), a Vietnamese dependency treebank comprising more than 10,000 sentences. Each sentence in this collection is meticulously annotated with dependency tree structures, providing a valuable resource for analyzing the grammatical relationships within Vietnamese texts.

Taking this legacy further, the **VLSP2020-UD** (Linh et al., 2020) corpus represents a significant advancement. As version 2 of the earlier collection, this dataset features over 9000 Vietnamese sentences and introduces substantial improvements, including the addition of more than 6000 new sentences compared to its predecessor. These enhancements make VLSP2020-UD an even more comprehensive resource for Vietnamese syntactic analysis.

Collectively, **VnDT** and **VLSP2020-UD** do more than simply document Vietnamese syntax—they chart the evolution of linguistic resources in Vietnam. From the detailed dependency structures of VnDT to the expanded and refined dataset of VLSP2020-UD, these treebanks not only deepen our understanding of Vietnamese grammar but also open new possibilities for advanced research and real-world language technology. Each release marks a step forward, reflecting the community's dedication to capturing the richness and complexity of Vietnamese through ever more comprehensive and sophisticated resources.

Semantic parsing involves analyzing the structure of the sentence and identifying its key components, including entities, relations, and actions. This involves identifying the words in the sentence and their respective parts of speech, as well as identifying the relationships between the words in the sentence. Once these components have been identified, they can be mapped to a structured meaning representation, such as a knowledge graph (Kang et al., 2026) or a semantic network. Semantic

Table 5
Comparison of modeling paradigms for Vietnamese dependency parsing.

Paradigm		Advantages	Limitations
Transition-based hand features	+	Linear-time parsing; interpretable; low memory	Error propagation in long sentences; high feature engineering cost
Graph-based neural		Models global tree structure; no error propagation; strong on long-range dependencies	Quadratic attention cost; requires larger treebank
PhoBERT head	+ biaffine	Strong contextual representations; best Vietnamese parsing accuracy; implicit syntactic knowledge from pretraining	RDRSegmenter preprocessing dependency; small Vietnamese treebank limits performance ceiling
LLM zero-shot		No treebank annotation required; generalizes across domains	Low structural accuracy; no canonical Vietnamese parsing evaluation for LLMs

parsing in Vietnamese is an active area of research, and there are several ongoing efforts to improve the accuracy and efficiency of the existing approaches. As the volume of Vietnamese text data continues to grow, semantic parsing is likely to become increasingly important for a wide range of natural language processing applications. In general, semantic parsing can be represented as a function that maps a sentence or phrase to a logical form. This function takes the form:

$$f : S \rightarrow L \quad (5)$$

where S is the space of all possible sentences or phrases, and L is the space of all possible logical forms. The function f maps each input sentence or phrase to the corresponding logical form.

As the volume of Vietnamese text data continues to expand, the demand for advanced semantic parsing tools tailored to the language's unique characteristics has grown more urgent, paving the way for innovative resources like the ViText2SQL dataset. ViText2SQL (Nguyen et al., 2020a) is a dataset that includes roughly 10,000 pairs of questions and SQL queries for the Text-to-SQL semantic parsing task. For ViText2SQL, researchers worked with the publicly available portion of Spider, consisting of 9691 questions and 5263 SQL queries over 166 databases, carefully adapting it to suit the Vietnamese language. This translation was far from a simple word-for-word conversion; it demanded a deep understanding of Vietnamese syntax, word order, and cultural nuances to ensure that each question and query remained both accurate and natural.

2.1.5. Relation extraction

The aim of **Relation Extraction (RE)** is to discover and determine the semantic relations between pairs of named entity mentions in the text. Formally, given a (set of) document(s), a desired RE system attempts to extract all pre-defined relationships specified in these documents by identifying the corresponding entities and defining the type of relationship between each pair of entities. This process can be represented as extracting relational triples (e_1, r, e_2) , where e_1 and e_2 are named entities, and r is the semantic relation connecting them. The successful identification of such relations is crucial for a wide range of downstream applications, including knowledge base construction, question answering, and information retrieval.

Despite significant progress in RE for popular languages such as English and Chinese, evaluations and research studies on this task for Vietnamese text are still in the early stages, with only a few initial achievements. The limited availability of high-quality annotated datasets and language-specific tools has posed substantial challenges to the advancement of Vietnamese RE. As a result, the development of standardized benchmarks and resources is essential to foster progress in this area.

Methodological Evolution

Vietnamese Relation Extraction (RE) remains one of the least developed areas of Vietnamese NLP, a direct consequence of the scarcity of annotated relational data. The sole publicly available Vietnamese RE dataset VLSP2020-RelEx was released only in 2020, meaning that the field lacks the decade-long benchmark history of NER or POS tagging. Early approaches to Vietnamese RE applied pipeline systems that

possibility for silicon sampling?

first performed NER to identify entity mentions, then applied pattern-based or SVM classifiers to classify relation types between entity pairs. More recent work has adapted cross-lingual pretrained models (XLM-R, PhoBERT) to RE via entity-marker finetuning, achieving promising results on VLSP2020-RelEx but remaining far below English RE benchmarks due to the small training set size (5,900 instances). The most critical bottleneck for Vietnamese RE is not modeling architecture but data scarcity: with only one benchmark dataset, generalisation claims remain difficult to validate. Table 6 summarises current approaches.

One notable contribution towards addressing this gap is the introduction of the **VLSP2020-RelEx** (Mai et al., 2020). This curated dataset consists of 1056 documents and 5900 instances of semantic relations, which were collected from Vietnamese news articles spanning various domains, including Location, Organization, and Person. Importantly, the dataset excludes all semantic relations between nominals other than the aforementioned entity types, thereby focusing on relations that are of general and practical interest. The authors selected four types of relation categories based on these three annotated entities, ensuring broad coverage and relevance for real-world applications.

2.1.6. Semantic similarity and relatedness

Semantic Similarity and Relatedness (SRS) quantifies how close in meaning two linguistic units (e.g., words, phrases, or sentences) are. This is formalized as a function assigning a real-valued score to a pair of texts:

$$\text{sim} : (X \times X) \rightarrow \mathbb{R} \quad (6)$$

where $\text{sim}(x, y)$ is the score indicating semantic similarity or relatedness between x and y . Given a reference unit x and a set of candidates $C = \{c_1, c_2, \dots, c_k\}$, the system outputs:

$$\{\text{sim}(x, c_i)\}_{i=1}^k \quad (7)$$

where each $\text{sim}(x, c_i)$ quantifies the semantic similarity or relatedness between x and c_i .

Methodological Evolution

Research on Vietnamese Semantic Similarity and Relatedness (SRS) remains one of the least developed areas of Vietnamese NLP, with only two publicly available benchmarks ViCon and ViSim-400 both released in 2018. Early approaches applied static word embeddings (Word2Vec, fastText) trained on Vietnamese corpora, measuring similarity via cosine distance between word vectors. While efficient, these methods conflated tonally distinct Vietnamese words that share the same Latin character sequence when diacritics are dropped, producing unreliable similarity scores for social media text. Contextual embedding models based on PhoBERT substantially improved semantic similarity scoring by conditioning representations on surrounding context, enabling discrimination of tonal synonyms and antonyms. A critical unresolved gap is the absence of sentence-level SRS benchmarks for Vietnamese: ViCon and ViSim-400 evaluate only word-pair similarity, leaving sentence and paragraph similarity — essential for downstream retrieval and QA tasks — entirely unevaluated. Table 7 summarises the approaches.

Table 6
Comparison of modeling approaches for Vietnamese relation extraction.

Approach	Advantages	Limitations	Data Need
Pattern-based / SVM	Interpretable; minimal labeled data; fast to develop with domain expertise	Low recall; brittle to syntactic variation; cannot generalize to unseen relation types	Low
Pipeline NER + classifier	Modular design; reuses existing Vietnamese NER models	Error propagation from NER stage; relation labels depend on NER quality	Moderate
End-to-end joint model	No error propagation; unified training; jointly learns entity and relation representations	Requires substantially more annotated data; architecturally more complex	High
PhoBERT + entity markers	Best current Vietnamese RE performance; transfer learning from Vietnamese pretraining	Only one Vietnamese RE benchmark (VLSP2020-RelEx, 5900 instances); small training set limits generalization	Moderate

Table 7
Comparison of modeling paradigms for Vietnamese semantic similarity and relatedness.

Paradigm	Advantages	Limitations
Word2Vec / fastText cosine	Efficient; no labeled data; captures distributional similarity	No contextual disambiguation; tonal Vietnamese words with different meanings share embeddings
Siamese BiLSTM	Sentence-level similarity; handles word-order differences	No pretraining; limited by Vietnamese training data size
PhoBERT sentence embeddings	Rich contextual representations; fine-grained similarity scoring; strong on ViCon and ViSim-400	Requires finetuning for sentence-level tasks; limited Vietnamese STS datasets for supervised training
Cross-lingual embeddings (LaBSE)	Strong multilingual similarity; no Vietnamese finetuning required	Not optimised for Vietnamese tonal synonym/antonym distinction

Despite its importance in natural language processing, Vietnamese resources for evaluating SRS remain limited. To address this, two notable datasets have been introduced. **ViCon** (Nguyen et al., 2018a) contains 1400 word pairs, divided into adjectives, nouns, and verbs, with equal numbers of synonym and antonym pairs in each category. This structure helps researchers distinguish between semantic similarity and dissimilarity in Vietnamese.

Additionally, **ViSim-400** (Nguyen et al., 2018a) includes 400 word pairs sampled from nouns, verbs, and adjectives, annotated for five semantic relations: synonymy, antonymy, hypernymy, co-hyponymy, and meronymy. Unlike ViCon, ViSim-400 is designed to reflect a continuum between similarity and relatedness, offering a more nuanced benchmark for semantic modeling.

Together, **ViCon** and **ViSim-400** provide valuable resources for developing and evaluating semantic similarity models in Vietnamese, though further expansion of such datasets is needed to advance the field.

2.1.7. Natural language inference

Natural language Inference (NLI) in Vietnamese refers to the task of determining the logical relationship between two pieces of text, such as a premise and a hypothesis. In this task, the system receives a premise as input, along with a hypothesis, and is required to classify the relationship between them as either entailment, contradiction, or neutral. NLI (Huynh et al., 2022; Van Huynh et al., 2026) is a challenging task for natural language processing systems, as it requires the system to have a deep understanding of the meaning and context of the text. The goal is to determine if the hypothesis is entailed, contradicted, or neutral with respect to the premise. The following formula shows how an NLI model can be represented:

$$\hat{y} = f_{NLI}(p, h), \quad (8)$$

where $\hat{y}$ is the predicted label, f_{NLI} is the NLI model, p is the premise sentence, and h is the hypothesis sentence. The model takes the pair of sentences as input and outputs a label indicating the relationship between them. Commonly used labels include “entailment”, “contradiction”, and “neutral”.

Methodological Evolution

Research on Vietnamese Natural Language Inference (NLI) is comparatively recent, with the first dedicated Vietnamese NLI dataset

(vnNLI-VLSP2021) released only in 2021. Early approaches relied on cross-lingual transfer from English NLI models (XLM-R, mBERT), which provided a zero-shot baseline but suffered from a significant transfer gap caused by the structural difference between English and Vietnamese entailment patterns, particularly the reliance on tonal and particle-based negation in Vietnamese. The release of ViNLI and VnNewsNLI provided native Vietnamese NLI training data, enabling PhoBERT finetuning to substantially outperform cross-lingual transfer baselines. The most recent dataset, ViANLI, introduces adversarially constructed examples specifically designed to expose model weaknesses in Vietnamese pragmatic inference, pushing evaluation beyond surface-level lexical matching. Table 8 compares the paradigms.

Given the complexity of NLI, high-quality datasets are essential for advancing Vietnamese language models. **vnNLI - VLSP 2021** (Quyen et al., 2022) was the first dataset and collected from reputable Vietnamese and English news websites, specifically from the health category. The dataset includes 16,185 sentence pairs for training and 4177 sentence pairs for testing. The main objective of this dataset is to facilitate the determination of the semantic relationship between a given pair of sentences. This semantic relationship can be categorized as either agree, disagree, or neutral/irrelevant. This dataset can be used to develop and evaluate natural language processing models for various applications, such as sentiment analysis and information retrieval, within the health domain.

Meanwhile, **ViNLI** (Huynh et al., 2022) is the Vietnamese dataset in the field of Natural Language Inference (NLI). It is an open-domain corpus designed to evaluate Vietnamese NLI models, and has been created and assessed using a rigorous quality control process. ViNLI consists of over 30,000 sentence pairs of human-annotated premises and hypotheses extracted from more than 800 online news articles covering 13 distinct topics. These datasets collectively mark significant strides in enhancing NLI research for Vietnamese.

VnNewsNLI (Nguyen & Nguyen, 2022) addresses critical limitations of machine-translated NLI datasets for Vietnamese by constructing a native-language corpus that eliminates cue marks and preserves authentic writing style. The researchers employed a semi-automatic methodology that leverages naturally occurring entailment pairs from VnExpress.net news articles, where titles typically paraphrase or summarize introductory sentences, thereby reducing annotation costs while ensuring linguistic authenticity. Annotators manually generated

Table 8
Comparison of modeling paradigms for Vietnamese natural language inference.

Paradigm	Advantages	Limitations
Feature-based SVM	Low data requirement; interpretable feature importance	No deep semantic modeling; brittle to lexical variation; fails on pragmatic inference
Cross-lingual transfer (XLM-R)	Zero-shot NLI from English; no Vietnamese NLI data required	Transfer gap between English and Vietnamese pragmatics; lower accuracy than finetuned models
PhoBERT finetune	Strong Vietnamese NLI accuracy; captures Vietnamese entailment patterns; benefits from Vietnamese-specific pretraining	Requires Vietnamese NLI training data; RDRSegmenter preprocessing dependency
LLM prompting	Zero-shot chain-of-thought reasoning; handles multi-sentence inference	Inconsistent on fine-grained contradiction detection; no canonical Vietnamese NLI benchmark for LLM evaluation

contradiction sentences using three systematic approaches: affirmative-negative transformations, structural negation patterns, and antonym substitution. The resulting dataset comprises 32,124 development samples and 11,878 test samples with balanced distributions across entailment, neutral, and contradiction categories.

2.2. NLP applications

2.2.1. Text classification

Text classification is a common task in natural language processing, and can be applied to many languages including Vietnamese. The goal of text classification is to automatically classify a text document into one or more predefined categories based on the content of the text. In Vietnamese text classification, some common categories include news topics such as politics, sports, and entertainment, as well as sentiment analysis tasks to classify whether a text expresses a positive or negative sentiment. Given an input text sequence $x = (x_1, x_2, \dots, x_n)$, a text classification model aims to predict its label y from a pre-defined set of labels Y . Let $f_\theta(x)$ represent the output of the model with parameters θ , the predicted label $\hat{y}$ can be obtained by:

$$\hat{y} = \operatorname{argmax}_y \in Y f_\theta(x)_y \quad (9)$$

where $f_\theta(x)_y$ is the score of the input sequence x being assigned with label y .

Methodological Evolution

Vietnamese text classification has followed the global NLP transition from feature-engineered statistical models to pretrained Transformer architectures. Early systems applied Naive Bayes and SVM classifiers with TF-IDF and n-gram features, achieving reasonable accuracy on topic classification tasks such as UIT-ViON but failing to capture semantic nuance and tonal variation. Deep learning approaches using CNN and BiLSTM architectures improved performance by modeling local and sequential Vietnamese text patterns, though they remained limited by the absence of large-scale Vietnamese pretraining. PhoBERT finetuning now achieves state-of-the-art results across all Vietnamese classification benchmarks by encoding rich contextual representations that capture the tonal and morphological signals critical for Vietnamese document classification. An emerging challenge is multi-label classification on Vietnamese social media, where a single comment may simultaneously trigger hate speech, offensive language, and complaint category labels a setting where current single-label classification architectures underperform. Table 9 summarises the paradigms.

To facilitate research and development in Vietnamese text classification, several high-quality datasets have been introduced, each tailored to specific classification tasks. For instance, the **UIT-ViNames** dataset (To et al., 2021) provides more than 26,000 full names annotated with gender labels, supporting the task of gender prediction based on Vietnamese names. This dataset, collected from various universities in Vietnam, ensures privacy by including only names and gender annotations (Male and Female). In contrast, the **UIT-ViON** dataset (Vietnamese Online Newspaper) (Tran et al., 2021b) addresses the need for large-scale,

multi-label text classification in the Vietnamese language. Comprising 260,000 annotated textual data points sourced from reputable online newspapers, UIT-ViON enables robust experimentation with machine learning models for short text classification across diverse topics. Together, these datasets exemplify the breadth of applications and the growing resources available for advancing Vietnamese text classification research.

2.2.2. Sentiment analysis

In the case of Vietnamese, **sentiment analysis** involves analyzing Vietnamese text to determine whether it expresses a positive, negative, or neutral sentiment. This task is particularly challenging due to the language's rich morphology, tonal variations, and diverse contextual expressions. As research in Vietnamese sentiment analysis has advanced, a variety of benchmark datasets have been introduced, each designed to address specific aspects of sentiment and emotion recognition.

Methodological Evolution

Vietnamese sentiment analysis has progressed through three generations of modeling approaches, mirroring the broader trajectory of NLP but with Vietnamese-specific inflection points. Lexicon-based methods formed the first generation, relying on manually curated Vietnamese sentiment lexicons and negation rules to classify document polarity without any labeled training data. While transparent and requiring no annotation, these systems failed to capture contextual sentiment inversion and the rich Vietnamese sarcasm patterns common in social media. Machine learning approaches particularly SVM with n-gram features substantially improved performance on the VLSP 2016-SA benchmark by learning data-driven lexical associations. The third generation, initiated by PhoBERT finetuning, delivers the current state-of-the-art across all Vietnamese SA benchmarks by encoding nuanced tonal and contextual sentiment signals that surface-form features miss. A notable Vietnamese-specific challenge is aspect-based sentiment analysis (ABSA): Vietnamese ABSA datasets such as UIT-ViSFD and UIT-ViSD4SA reveal that aspect boundary detection is significantly harder in Vietnamese than in English due to the absence of morphological aspect markers. Table 10 compares the paradigms.

UIT-VSFC (Nguyen et al., 2018b) is a benchmark dataset including feedback collected from students at a Vietnamese university through surveys administered by the university at the end of each semester between 2013 and 2016. The dataset comprised more than 16,000 sentences, providing a valuable resource for general sentiment classification tasks.

UIT-VSMEC (Ho et al., 2020) was created in 2019 by Ho et al. as the Vietnamese Social Media Emotion Corpus. This dataset consists of 6927 sentences annotated by humans with six emotion labels, contributing to research on fine-grained emotion recognition in the Vietnamese language.

To address aspect-based sentiment analysis (ABSA), **UIT-ViSFD** (Luc Phan et al., 2021) was developed as a benchmark corpus comprising 11,122 comments related to mobile e-commerce. The dataset features strict annotation schemes to ensure high-quality

Table 9
Comparison of modeling paradigms for Vietnamese text classification.

Paradigm	Advantages	Limitations
Naive Bayes / SVM	Extremely fast training; low data requirement; interpretable decision boundary	Bag-of-words assumption; no word order or tonal modeling; brittle to Vietnamese morphological variation
CNN / BiLSTM	Local and sequential feature capture; handles Vietnamese n-gram patterns	No pretraining; limited long-document modeling; sensitive to vocabulary coverage
PhoBERT finetune	SOTA on all Vietnamese classification benchmarks; contextual tonal and semantic modeling; minimal task-specific architecture required	Overkill for simple short-text tasks; RDRSeg-menter preprocessing dependency
LLM zero-shot / few-shot	No labeled data for new categories; flexible label schema; handles implicit Vietnamese class signals	Lower accuracy than finetuned models on fixed-label benchmarks; high inference cost

Table 10
Comparison of modeling paradigms for Vietnamese sentiment analysis.

Paradigm	Advantages	Limitations
Lexicon-based	No training data required; interpretable; zero-shot applicable to new domains	No contextual modeling; fails on sarcasm, negation, and tonal sentiment inversion
SVM + n-gram features	Data-driven; handles implicit sentiment; moderate labeled data requirement	Fixed feature templates; high-dimensional sparse features; no deep semantic encoding
BiLSTM / CNN	Contextual encoding; character-level Vietnamese morphology handling; end-to-end training	No pretraining; sensitive to data size and domain; no tonal signal modeling
PhoBERT / ViDeBERTa	SOTA across all SA benchmarks; captures tonal and contextual sentiment signals; strong ABSA performance with task-specific heads	High compute cost; degrades on diacritic-dropped informal text; ABSA requires custom output layers

aspect-based sentiment labels. Further, **UIT-ViSD4SA** (Thanh et al., 2021a) provides a benchmark Vietnamese smartphone feedback dataset for ABSA and span detection, consisting of 35,396 human-annotated spans on 11,122 feedback comments. Each comment is manually annotated according to its spans towards ten fine-grained aspect categories with sentiment polarities, and the dataset is split into training, development, and test sets.

VLSP2018-ASA (Nguyen et al., 2019b) further addresses ABSA in the Vietnamese language by providing two datasets comprising reviews in the restaurant and hotel domains. These datasets were created by manually annotating reviews from 150 hotels in each of the three cities of Ha Noi, Da Nang, and Ho Chi Minh City, resulting in 4751 reviews for the restaurant domain and 5600 reviews for the hotel domain.

For document-level sentiment classification, **VLSP2016-SA** (Nguyen et al., 2019b) involves classifying Vietnamese reviews or documents into one of four labels: POS (positive), NEG (negative), NEU (neutral), and USELESS. The data domain consists of technical article reviews collected from three source sites: tinhte.vn, vnexpress.net, and Facebook, with a total of 12,196 comments.

In addition to sentiment and aspect-based analysis, recent research has focused on hate speech and offensive language detection in Vietnamese social media. **UIT-ViHSD** (Luu et al., 2021b) is a human-annotated dataset designed for automatically detecting hate speech on social networks, consisting of over 30,000 comments labeled as CLEAN, OFFENSIVE, or HATE. **UIT-ViHOS** (Hoang et al., 2023) is the first human-annotated corpus for hate and offensive spans in Vietnamese comments, including 26,000 spans across 11,000 comments, along with definitions of hateful and offensive spans and detailed annotation guidelines. Similarly, **VLSP 2019 HateSpeechDetection** (Vu et al., 2020) aims to detect the presence of hate speech in Vietnamese textual items extracted from Facebook, involving multi-class classification with three classes: “HATE”, “OFFENSIVE”, and “CLEAN”, and comprising 25,431 textual items.

Another important contribution is **ViCTSD** (Nguyen et al., 2021a), the **UIT-ViCTSD** (Vietnamese Constructive and Toxic Speech Detection) dataset, which was created to support constructive and toxic speech detection in Vietnamese. It includes 10,000 comments labeled

as Constructive or Non-Constructive, with additional labels for four levels of toxicity: Very toxic, Toxic, Quite toxic, and Non-toxic.

Finally, **UIT-ViOCD** (Nguyen et al., 2021c) (Open-domain Complaint Detection) is a collection of 5485 human-annotated reviews sourced from e-commerce sites and the Google Play Store. The reviews are categorized into four areas: fashion, cosmetics, applications, and phones, supporting research on open-domain complaint detection in Vietnamese product reviews.

Collectively, these datasets form a comprehensive ecosystem for Vietnamese sentiment analysis, spanning from general sentiment and emotion recognition to aspect-based sentiment analysis and hate speech detection. By providing high-quality, annotated data, they enable the development and evaluation of advanced machine learning models, thereby driving progress in Vietnamese natural language processing research.

2.2.3. Question answering and machine reading comprehension

Given a question q and a context c , the goal of QA is to find the answer a such that:

$$a = \operatorname{argmax}_a P(a|q, c) \quad (10)$$

In other words, the answer a that maximizes the probability of a given the question q and context c is selected as the predicted answer. This can be computed using various models, such as neural networks or rule-based approaches, which estimate the conditional probability of a given q and c .

Methodological Evolution

Vietnamese Machine Reading Comprehension (MRC) has evolved significantly from a low-resource field to one rich with diverse benchmarks and advanced modeling techniques. Early approaches faced a scarcity of high-quality datasets, leading to the creation of foundational resources like ViMMRC and UIT-ViQuAD, which established essential baselines using lexical-matching and early deep learning methods. These systems primarily focused on explicit, answerable questions within limited domains and could not handle unanswerable questions or multi-hop reasoning. The introduction of PhoBERT finetuning represented a

Table 11

Comparison of modeling paradigms for Vietnamese question answering and machine reading comprehension.

Paradigm	Advantages	Limitations
Rule-based / keyword matching	No training data; fast retrieval; high precision on exact-match questions	No comprehension; fails on paraphrased or implicit questions; no span extraction capability
BiLSTM + attention	Contextual span extraction; handles answer boundary detection	Fixed context window; limited multi-hop reasoning; requires task-specific architecture
PhoBERT finetune	Strong extractive MRC performance; handles unanswerable questions (UIT-ViQuAD 2.0); rich Vietnamese contextual representations	Extractive only; cannot generate abstractive answers; RDRSegmenter dependency
CafeBERT finetune	Current SOTA 92.47 F1 on UIT-ViQuAD 2.0; Vietnamese-optimised tokenizer	Limited reproducibility due to undisclosed training methodology and absence of publicly available model checkpoints
Generative LLM (RAG)	Open-domain QA capability; abstractive answer generation; no fixed passage required	Hallucination risk on Vietnamese factual queries; no standardized evaluation suite

decisive shift, enabling span extraction models that encode long-range contextual dependencies within Vietnamese passages. UIT-ViQuAD 2.0 further advanced the field by incorporating 11,501 unanswerable questions, forcing models to distinguish absent information rather than always extracting a span — directly reducing hallucination in open-domain QA systems. The most recent frontier is generative QA via LLMs with Retrieval-Augmented Generation (RAG), which removes the fixed-passage constraint and enables open-domain Vietnamese question answering. Table 11 compares these paradigms.

Textual-based QA is a type of question answering system that uses natural language text as input and provides relevant answers based on the content of the text. It can be divided into several subcategories, such as *knowledge-based QA (KBQA)* and *Machine Reading Comprehension (MRC)*.

VIMQA (Le et al., 2022) is the new dataset for multi-hop question answering on Vietnamese text. The dataset is constructed on the Wikipedia domain which contains 10,000 human annotated question-answer pairs. VIMQA is a challenging dataset for evaluating both single-hop and multi-hop question answering systems.

Extending the scope of question answering beyond unstructured data to structured knowledge, Dao et al. (2025) presented **Open-ViTabQA**. Addressing the critical lack of benchmarks for structured data in low-resource languages, this dataset comprises 9911 question-answer pairs derived from 329 Wikipedia tables across 41 domains. Together with the visual datasets, Open-ViTabQA ensures that the Vietnamese NLP community has access to a comprehensive suite of benchmarks covering diverse modalities, from natural images and scene text to tabular data.

Knowledge-Based Question Answering (KBQA) is the task of answering natural language questions using structured knowledge bases. The task of KBQA involves mapping a natural language question to a structured query language such as SPARQL (Borotto & Ricca, 2023), which is used to retrieve relevant information from the knowledge base. The mapping function can be represented as:

$$f_{map}(q) = s \quad (11)$$

where q is the natural language question, s is the corresponding SPARQL query, and f_{map} is the mapping function that maps the question to the SPARQL query. The SPARQL query is then executed on the knowledge base, resulting in a set of candidate answers A .

$$A = f_{exec}(s) \quad (12)$$

where f_{exec} is the query execution function that executes the SPARQL query s on the knowledge base, and A is the set of candidate answers. The task of KBQA is to rank the candidate answers based on their relevance to the question. This can be done using a scoring function that measures the similarity between the question and each candidate answer. The scoring function can be represented as:

$$f_{score}(q, a_i) = s_i \quad (13)$$

where q is the natural language question, a_i is the i th candidate answer, s_i is the score assigned to the i th candidate answer, and f_{score} is the scoring function that computes the score for each candidate answer. The candidate answers can then be ranked based on their scores to obtain the final answer to the question.

COVID-KGQA (Tran et al., 2022a) is a bilingual dataset focused on COVID-19 designed to further enhance the diversity of domains and languages in the field of question answering. Beginning with the latest version of DBpedia, the authors developed a question-answer corpus containing over 1000 bilingual pairs. The COVID-KGQA dataset provides a valuable resource for training and evaluating question answering models focused on COVID-19 in both English and Vietnamese.

Machine Reading Comprehension (MRC) is the task of answering questions based on a given text passage. This task can be formalized as a function $f(p, q)$ that maps a passage p and a question q to an answer a , as shown below:

$$a = f(p, q) \quad (14)$$

where p is the text passage, q is the question, and a is the answer. The ultimate goal of MRC is to find the most accurate answer a for each pair (p, q) , typically by training a model on a large dataset of text passages, questions, and corresponding answers, and minimizing the loss between the predicted and ground-truth answers.

Vietnamese Machine Reading Comprehension (MRC) has evolved significantly from a low-resource field to one rich with diverse benchmarks and advanced modeling techniques. Initially, research faced a scarcity of high-quality datasets, leading to the creation of foundational resources like ViMMRC (Nguyen et al., 2020c), the first multiple-choice dataset based on elementary school texts, and UIT-ViQuAD (Nguyen et al., 2020b), a large-scale span-extraction dataset derived from Vietnamese Wikipedia. These early efforts established essential baselines, moving the field from simple lexical-based approaches to deep learning methods, though they primarily focused on explicit, answerable questions within limited domains.

Recent advancements have pushed the boundaries of Vietnamese MRC towards more complex and realistic cognitive tasks. To address the “answerability” problem, UIT-ViQuAD 2.0 (Kiet et al., 2022) was introduced as a substantial expansion of its predecessor, comprising 35,990 question-answer pairs derived from 176 Wikipedia articles. Crucially, it incorporates 11,501 unanswerable questions roughly one-third of the dataset generated through rigorous methods such as entity swapping and antonym substitution. This design forces models to discern when information is absent rather than simply extracting text spans, thereby reducing hallucinations in open-domain QA systems.

Concurrently, the ViMMRC 2.0 dataset (Luu et al., 2025) expanded the scope of multiple-choice testing to cover literature curricula from grades 1 through 12, requiring models to perform multi-hop reasoning and grasp implicit contexts. Unlike earlier datasets limited to primary school prose, ViMMRC 2.0 includes 699 passages featuring both prose and poetry, paired with 5273 questions. These questions are categorized

into diverse reasoning types, including Wh-questions, listed options, and Yes/No questions**, challenging models to understand the “macrostructure” of literary texts rather than just local propositions.

For domain-specific comprehension, especially in healthcare, **UIT-ViNewsQA** (Van Nguyen et al., 2022) and **ViHealthQA** (Nguyen et al., 2022c) provide valuable resources. Both corpora contain over 22,000 question-answer pairs derived from thousands of Vietnamese healthcare news articles, with answers carefully extracted as relevant text spans. Additionally, **ViCoQA** (Luu et al., 2021a) offers 10,000 questions and answers from 2000 health-related conversations based on news articles from VnExpress, one of Vietnam’s leading online newspapers.

Finally, **UIT-ViCoV19QA** (Thai et al., 2022) addresses the urgent need for reliable COVID-19 information by providing 4500 question-answer pairs sourced from trusted medical content. Each question in this dataset is paired with one or more paraphrased answers, supporting the development of robust question answering systems for public health. The Vietnamese MRC ecosystem is the best-resourced in this survey (15 datasets). Current best performance on UIT-ViQuAD 2.0 is 92.47 F1 by CafeBERT (Do et al., 2024b), representing a 17-point gap over mBERT (75.33 F1), confirming language-specific pre-training as the dominant performance driver.

Visual Question Answering (VQA) in Vietnamese is an emerging field within natural language processing that focuses on enabling machines to answer questions about images or videos using Vietnamese textual descriptions. The central goal is to train models that can understand the context of visual content and generate appropriate textual answers to user queries. Formally, the task is to determine the most probable answer A from a set of answer options O , given an image I and a question Q . This objective can be mathematically expressed as:

$$A = \underset{A \in O}{\operatorname{argmax}} P(A|I, Q, O), \quad (15)$$

where argmax denotes the function that returns the answer maximizing the conditional probability $P(A|I, Q, O)$. This probability is estimated by a model that jointly processes the image I and the question Q to generate a probability distribution over the answer options O .

The trajectory of research in Vietnamese Visual Question Answering (VQA) began with the necessity of establishing foundational resources for low-resource languages. The field was initiated with the introduction of **ViVQA** (Tran et al., 2021a), a dataset designed to enable initial model training and evaluation. ViVQA contains 10,328 images sourced from the MS COCO dataset (Lin et al., 2014), paired with 15,000 question-answer pairs. While this dataset ensured a wide variety of everyday scenes and objects, its reliance on Western-centric imagery limited its ability to capture the specific cultural nuances of Vietnam.

To bridge this cultural gap, the **OpenViVQA** (Nguyen et al., 2023f) dataset was developed as a significant milestone, representing the first comprehensive open-domain Vietnamese VQA dataset with open-ended answers. Building upon the foundation laid by ViVQA, this dataset includes over 11,000 images and more than 37,000 manually annotated question-answer pairs. Crucially, unlike its predecessor, the images in OpenViVQA were captured exclusively in Vietnam. This shift allows researchers to explore distinctively Vietnamese visual scenes, fostering the development of models that are not only linguistically accurate but also culturally context-aware.

As the field matured, the focus expanded from general scene understanding to rigorous cognitive evaluation. In 2024, **ViCLEVR** (Tran et al., 2024) was introduced to assess visual reasoning capabilities and address potential biases in AI research. Extending the diagnostic CLEVR framework to the Vietnamese context, ViCLEVR comprises over 26,000 images and 30,000 annotated pairs. Each question is carefully labeled to specify the type of reasoning involved, providing a robust platform for assessing complex cognitive skills and setting a new standard for unbiased, linguistically diverse VQA resources.

Concurrent with advancements in reasoning, researchers identified a limitation in existing datasets regarding the comprehension of embedded text within images. To address this, Van Nguyen et al. (2025)

introduced **ViTextVQA**, the first large-scale dataset specifically designed for Vietnamese scene text comprehension. With 16,762 images and 50,342 question-answer pairs collected from diverse Vietnamese street views, ViTextVQA challenges models to extract and reason about explicit information found on signboards, banners, and billboards, moving beyond simple object recognition to complex multimodal literacy.

2.2.4. Machine translation

Machine Translation in Vietnamese is the task of automatically translating text from one language to another, specifically from Vietnamese to another language or from another language to Vietnamese. This task involves training a machine translation system on a large parallel corpus of texts in both source and target languages and using statistical or neural models to generate translations. Machine translation in Vietnamese is a challenging task due to the complex morphology and syntax of the Vietnamese language, as well as the need to accurately capture the meaning and context of the text. Given a source language sentence $S = s_1, s_2, \dots, s_{n_s}$, the goal of machine translation is to generate a target language sentence $T = t_1, t_2, \dots, t_{n_t}$ such that T conveys the same meaning as S . The conditional probability of a target sentence T given a source sentence S can be expressed as:

$$P(T|S) = \prod_{i=1}^{n_t} P(t_i|t_{<i}S) \quad (16)$$

where $t_{<i}$ denotes the subsequence of the target sentence up to position $i - 1$. The joint probability of a source sentence and a target sentence can be expressed as:

$$P(S, T) = P(T|S)P(S) \quad (17)$$

where $P(S)$ is the probability of a source sentence. The goal of machine translation is to find the most likely target sentence given a source sentence, which can be expressed as:

$$\hat{T} = \underset{T}{\operatorname{argmax}} P(T|S) \quad (18)$$

where $\hat{T}$ is the translation of the source sentence S .

Methodological Evolution

Vietnamese machine translation (MT), particularly the English–Vietnamese pair, has been transformed by the availability of large parallel corpora and pretrained sequence-to-sequence models. Statistical MT systems based on phrase-based models were the dominant approach until approximately 2018, leveraging parallel news corpora but suffering from fluency failures on long-distance reordering a particularly acute problem for Vietnamese given its SOV-adjacent constituent order in subordinate clauses. Neural MT systems based on the standard Transformer architecture substantially improved fluency by modelling global source-target alignment through attention. The release of PhoMT a 3.02 million sentence pair corpus of high-quality English–Vietnamese parallel text provided the scale necessary to train competitive Vietnamese-specific neural MT systems. Sequence-to-sequence pretrained models (BARTpho, mBART-50) further improved translation quality by leveraging Vietnamese pretraining as a prior for the generation decoder. Table 12 compares the paradigms and Table 13 reports benchmark scores.

Table 13 demonstrates that pretrained sequence-to-sequence models consistently outperform the standard Transformer baseline trained from scratch, with BARTpho-word improving over the Transformer by 2.9 BLEU on Vi→En. The vinaï-translate system achieves the highest scores on both directions, suggesting that Vietnamese-specific finetuning strategies beyond standard pretraining remain an important research direction. The gap between current best BLEU (35.9) and typical English–German benchmarks (>40 BLEU) reflects the relative difficulty of the English–Vietnamese pair, driven by the large typological distance and tonal disambiguation requirements.

PhoMT (Doan et al., 2021) represents the first publicly available large-scale Vietnamese–English machine translation dataset, containing 3.02 million high-quality sentence pairs. Developed by VinAI Research in 2021, PhoMT addresses critical data scarcity that previously

Table 12
Comparison of modeling paradigms for Vietnamese machine translation.

Paradigm	Advantages	Limitations
Phrase-based SMT	Interpretable phrase table; controllable output; low computational cost	Long-distance reordering errors; fluency failures in Vietnamese output; manual feature engineering required
Standard Transformer	Strong global alignment via attention; fully parallelizable training	Requires large parallel corpus; no Vietnamese-specific prior; limited low-resource generalization
mBART-50 / BARTpho	Pretrained seq2seq prior; strong low-resource generalization; multilingual transfer benefits	Large model size; BARTpho-word requires RDRSegmenter preprocessing
vinai-translate	SOTA on PhoMT benchmark; Vietnamese-optimised finetuning strategy; strong on both translation directions	Closed architecture details; limited domain generalization beyond news

Table 13
Model performance on PhoMT benchmark (BLEU score). Best results are **bolded**.

Model	Vi→En BLEU	En→Vi BLEU
Phrase-based SMT	24.3	26.1
Standard Transformer	29.8	31.5
mBART-50	31.2	33.1
BARTpho-word	32.7	34.8
vinai-translate	34.2	35.9

hindered Vietnamese MT research, expanding beyond the limited 133K pairs in IWSLT15. The dataset's construction involved rigorous quality control through automatic filtering, manual verification, and diverse domain coverage from sources like WikiHow, TED Talks, and OpenSubtitles. PhoMT established comprehensive baseline evaluations using state-of-the-art models, revealing significant performance gaps and unique challenges in Vietnamese language processing. Its public release democratized access to Vietnamese translation data, enabling researchers worldwide to develop systems.

The most recent dataset in the field of Vietnamese-English medical machine translation is **MedEV** (Vo et al., 2024), a high-quality and large parallel corpus containing approximately 360,000 sentence pairs collected from 27,000 medical documents, including journal abstracts, manuals, theses, and expert translations. Developed through careful data collection, sentence segmentation, automatic and manual alignment, MedEV sets a new benchmark for domain-specific translation research by offering both scale and rigorous annotation quality with less than 0.5% misalignment in its validation and test sets. Its availability has enabled researchers to empirically show that fine-tuned medical MT models, such as vinai-translate, outperform general-purpose translation engines in this specialized domain, making MedEV an indispensable foundation for future work and development in Vietnamese-English medical language technologies. Vietnamese MT is anchored by PhoMT and MedEV. BLEU scores on PhoMT reach approximately 31.5 (Vi→En) and 29.8 (En→Vi) for standard Transformer models; BARTpho-based approaches (Tran et al., 2022b) report up to +3.2 BLEU through pre-training-augmented fine-tuning.

2.2.5. Summarization

Summarization is the process of creating a condensed version of a given text document while preserving its key information content. There are two main types of text summarization: extractive and abstractive. In extractive summarization, the summary is generated by selecting and combining the most important sentences or phrases from the source document. This approach is generally less demanding in terms of natural language processing and often produces fluent and coherent summaries, but it may lead to redundancy and lacks the creativity of human-written summaries. The process can be mathematically represented as:

$$S = \arg \max_{S'} \sum_{s \in S'} f(s) \quad (19)$$

where S is the summary, S' is a set of candidate sentences, and $f(s)$ is a scoring function that measures the importance or relevance of each sentence.

In *extractive summarization*, the summary is created by selecting and combining the most important sentences or phrases from the source document. This approach requires less natural language processing and can generate more fluent and coherent summaries, but may result in redundancy and lack of creativity.

Abstractive summarization, on the other hand, involves generating new sentences that capture the essential meaning of the source text in a concise and coherent manner. This method requires a deeper understanding of the text and more advanced natural language generation capabilities. While abstractive summarization can produce more informative and creative summaries, it is also more prone to grammatical errors and issues with fluency, especially in languages with complex structures like Vietnamese. The variability and richness of Vietnamese present additional challenges for the development of high-quality summarization models.

Methodological Evolution

Vietnamese text summarization has evolved from extractive graph-based methods to abstractive neural generation, following the global trajectory of summarization research while being constrained by Vietnamese-specific data scarcity. Early extractive systems applied TextRank and LexRank graph algorithms to Vietnamese news, selecting salient sentences based on inter-sentence similarity without generating new text. These systems required no labeled data and served as practical baselines, but their outputs lacked coherence and missed domain-specific Vietnamese discourse conventions. Sequence-to-sequence neural systems with attention mechanisms improved coherence by generating abstractive summaries, though they required parallel document-summary training data that was scarce for Vietnamese until the release of VnDS and ViMs. BARTpho (Tran et al., 2022c) represents the current state-of-the-art for Vietnamese abstractive summarization, leveraging Vietnamese-specific denoising pretraining to generate fluent summaries with minimal finetuning data. Table 14 summarises the methodological progression.

To support research in Vietnamese text summarization, several datasets have been introduced. **The Vietnamese News Dataset (VNDS)** (Nguyen et al., 2019c) was created by collecting articles from major news sources such as tuoitre.vn, vnexpress.net, and nguoiduatin.vn, spanning topics like world news, law, and business from 2016 to 2019. This dataset contains more than 150,000 sentences, making it a valuable resource for training and evaluating summarization models. Another important resource is **ViMs corpus** (Tran et al., 2020), which includes 1945 documents and 25,100 sentences collected from various popular Vietnamese news websites. The corpus is organized into 300 topics, with each topic containing several related documents, thus supporting research on multi-document summarization.

Additionally, **VSoLSCSum dataset** (Nguyen et al., 2016) focuses on summarizing Vietnamese social life events. It was constructed from

Table 14
Comparison of modeling paradigms for Vietnamese text summarization.

Paradigm	Advantages	Limitations
Extractive (TextRank / LexRank)	No training data required; fast; preserves original wording; fully interpretable	Incoherent multi-sentence output; misses cross-sentence information fusion; no Vietnamese discourse modeling
Seq2Seq + attention	Abstractive fluency; learns Vietnamese compression patterns; handles paraphrase and re-formulation	Requires parallel document-summary data; repetition and hallucination failures on long Vietnamese documents
BARTpho finetuning	SOTA abstractive quality on VnDS and ViMs; Vietnamese denoising pretraining prior; syllable variant avoids preprocessing	Large model size; BARTpho-word requires RDRSegmenter; limited evaluation on non-news Vietnamese domains
LLM prompting	Zero-shot summarization capability; handles diverse document structures; no finetuning required	Verbosity and hallucination tendency; no standardized Vietnamese summarization benchmark for LLMs

141 web documents related to 12 special events, resulting in 3760 sentences, 2448 gold-standard reference summaries, and 6926 comments. This dataset is particularly useful for studying summarization in the context of social events and user-generated content. Current state-of-the-art ROUGE-2 on VNDS is ≈ 38.4 achieved by BARTpho-word (Tran et al., 2022b), a +5.1 ROUGE-2 improvement over PhoBERT-based extractive baselines (Nguyen & Tuan Nguyen, 2020).

2.2.6. Legal engineering

Legal Engineering is a specific domain adaptation that employs NLP techniques, such as text embedding, language modeling, and downstream tasks like text retrieval and question answering, to build an intelligent system in the Legal domain. Because legal texts are complex due to formal language and specialized terminology, they require the computer's ability to acquire in-depth knowledge and reason about complex legal concepts and principles (Anh et al., 2023). Practically, Legal Engineering consists of two main tasks:

- **Legal Retrieval:** The Legal Retrieval task requires the computer to provide relevant documents or legal articles for an input query. Particularly, the task is denoted as: given a legal query q and a legal database DB , the output of the model should be a list of $D = d_1, \dots, d_n$ containing n relevant documents to q ($D \in DB$). In practice, D can be at the document or article level depending on the specific task. Most legal retrieval tasks in Vietnamese, such as the ZaloAI Challenge - Legal Retrieval² and ALQAC series (Do et al., 2025a, 2024a; Nguyen et al., 2022a, 2023a; Thanh et al., 2021b), require the model to retrieve relevant articles for a query across multiple legal documents, meaning the model needs to identify the correct articles for a given query. Despite many attempts to develop state-of-the-art techniques such as robust contextual embedding, the Legal Retrieval task still poses a challenge in the correctness of retrieved candidates, especially as legal articles become longer and more complex in structure. Obtained from the 2024 ALQAC competition (Do et al., 2024a), the state-of-the-art approaches for Legal Retrieval are multi-stage, which first identify the top-k relevant document candidates using a lexicon retriever such as BM25. Then, the retrieval model aligns the query with articles in each document using a semantic embedding language model such as VnLawBERT (Chau et al., 2020), ViT5 (Phan et al., 2022), or even Large Language Models (LLMs) to obtain the final article candidates.
- **Legal Question Answering:** This task aims to provide the correct answers for the given legal query. Notably, the task is denoted as: given a legal query q and the list of n relevant documents $D = d_1, \dots, d_n$, the goal is to provide the correct answer A to the users. Commonly, the answer to a legal question is a piece of sentence, or a paragraph extracted directly from the relevant articles or documents, which is called *extractive* answer. However, the intelligent Legal Engineering

systems require the answers to be correct, coherent, and comprehensible to the users. Hence, the answers need to be rewritten or paraphrased from the original retrieved candidate to generate the final answer, which is called *abstractive* answers. In the era of Large Language Models, Legal Question Answering achieved high performance on the legal question-answering task. However, since Legal Question Answering depends on the correctness of Legal Retrieval, its performance will be affected by the accuracy of the retrieval steps. Additionally, most current Vietnamese legal QA datasets are constructed from gold-retrieval documents, meaning the relevant document set D is determined by experts; there is still room for the development of a fully automatic QA pipeline for legal domains.

To support research on the Vietnamese legal engineering task, several approaches to constructing a high-quality dataset are proposed. Unlike other domains, a legal dataset requires expert involvement in data creation to ensure its reliability and accuracy. First, the ALQAC series (Do et al., 2024a; Nguyen et al., 2022a, 2023a; Thanh et al., 2021b) provides the pilot datasets for the Vietnamese legal question-answering (QA) competition, which was entirely constructed by humans and verified by lawyers. Second, several datasets are built for the Legal QA task, such as R2GQA (Do et al., 2025b), which covers legal regulations in higher education, ViBidLQA (Ha et al., 2024) on bidding law, and the Vietnamese Labor Law QA dataset (Pham et al., 2024), which supports question-answering about Labor Laws. Besides, the VLSP2025-MLQA-TSR (Son et al., 2025) provides a new multimodal dataset for legal question answering. More details about the current datasets for Vietnamese Legal Engineering can be found in Table 17.

The annual ALQAC shared task series (2021–2025), supplemented by ViBidLQA, VNLaborLawQA, R2GQA, and VLQA, comprises $\approx 38,000$ queries across Vietnamese statutory, civil, and administrative law. Despite this breadth, the absence of legal instruction-tuning corpora means free-text generation on legal documents remains of low quality making this a high-priority target for domain-specific SLM development (Section 4).

Methodological Evolution

Vietnamese Legal NLP is a rapidly emerging sub-field, catalyzed primarily by the annual ALQAC shared task series (2021–2025). Early systems applied BM25 keyword retrieval to match user queries against statutory article databases, achieving high precision on exact legal citations but completely failing on paraphrased or implicitly reasoned legal questions. Subsequent work introduced BERT-based bi-encoder and cross-encoder re-ranking pipelines, substantially improving recall on semantic legal retrieval tasks. PhoBERT finetuning on legal corpora further improved performance by adapting Vietnamese representations to the formal, archaic vocabulary characteristic of Vietnamese statutory text, which differs markedly from the news-domain text on which PhoBERT was pretrained. The most recent frontier is Retrieval-Augmented Generation (RAG) for multi-article legal QA, as demon-

² <https://challenge.zalo.ai>

Table 15
Comparison of modeling paradigms for Vietnamese legal NLP.

Paradigm	Advantages	Limitations
Rule-based legal matching	Fully interpretable; high precision on exact statutory citations; no training data required	Cannot handle paraphrased queries; brittle to legislative updates; zero recall on implicit legal reasoning
BM25 + re-ranking	Fast retrieval; handles large Vietnamese legal corpora; no annotation required	Lexical matching only; misses semantic legal equivalences; struggles with archaic Vietnamese legal vocabulary
PhoBERT finetune	Strong semantic legal retrieval; handles Vietnamese legal terminology; robust to query paraphrase variation	Pretrained on general text; limited coverage of Vietnamese legal-specific terminology; RDRSegmenter dependency
LLM + RAG	Open-domain legal QA capability; handles multi-article reasoning; abstractive answer generation	Hallucination risk on specific Vietnamese statutory facts; no standardized legal LLM evaluation suite

Table 16
Comparison of modeling paradigms for Vietnamese fact checking.

Paradigm	Advantages	Limitations
Evidence retrieval + SVM	Modular; interpretable; no end-to-end training required	Error propagation from retrieval stage; no deep semantic verification
NLI-based pipeline	Reuses strong Vietnamese NLI models; claim-evidence entailment framing is linguistically principled	Requires high-quality evidence retrieval; NLI models not trained on Vietnamese misinformation patterns
PhoBERT end-to-end	Joint retrieval and verification; Vietnamese-specific representations; strong on ViWikiFC benchmark	Limited to Wikipedia-domain evidence; does not handle numerical or temporal claim verification
LLM chain-of-thought	Multi-hop reasoning capability; handles complex Vietnamese claim structures; applicable to ViNumFCR numerical facts	High hallucination risk; no standardized Vietnamese fact-checking leaderboard for LLMs

strated by R2GQA, which requires models to synthesize reasoning across multiple retrieved statutory passages to generate a coherent legal answer. Table 15 summarises the approaches.

2.2.7. Fact-checking task

Fact-checking is the process of assessing whether a claim, whether written or spoken, is correct (Guo et al., 2022). This is another application of NLP in social domains that aims to reduce the negative impact of misleading information on social media. In Vietnam, the popularity of social media makes Fact-checking tasks more crucial to curb the spread of deceptive and fabricated news, especially during important events. However, the Fact-checking task is challenging because it requires multiple steps to verify the information, including claim detection, retrieving relevant information, verifying the claim, and finally explaining the justification for the veracity (Kotonya & Toni, 2020). The enormous size of information on the Internet also indicates a challenge for the Fact-checking task.

Methodological Evolution

Vietnamese fact checking as a formal NLP task emerged relatively recently, driven by growing concern over misinformation on Vietnamese social media. The ReINTEL dataset was the first Vietnamese resource specifically targeting unreliable information detection, framing the problem as a binary classification task on social media posts. Subsequent work expanded the scope to evidence-based verification, requiring systems to retrieve supporting or refuting evidence from external sources before making a veracity judgment — an approach formalized in ViWikiFC. The most recent datasets, ViFactCheck and ViNumFCR, introduce domain-specific challenges: ViFactCheck targets multi-domain Vietnamese claims requiring document-level evidence fusion, while ViNumFCR specifically targets numerical and statistical claim verification, a sub-task where current Vietnamese NLP models exhibit the largest performance gap relative to English benchmarks. Table 16 summarises the paradigms.

The ViWikiFC (Le et al., 2024) is one of the first attempts in constructing a large-scale and high-quality dataset in the Vietnamese language. Next, the ViFactCheck (Hoa et al., 2025) and ViNumFCR (Luong

et al., 2025) are two new challenge datasets for fact-checking that feature multi-hop reasoning and complex evidence representation. Moreover, the Fact-checking system combines different NLP techniques, such as information retrieval and NLI for retrieving evidence, multi-stage and multimodal classification for verifying truthfulness, and generative models for constructing explanations. To conclude, the Fact-checking task is a new and challenging initiative in Vietnam, aimed at developing a Fact-checking system for practical use.

Resources include ReINTEL, ViWikiFC, ViFactCheck, and ViNumFCR, with progressive inclusion of numerical reasoning (ViNumFCR) and multi-hop evidence chains reflecting community maturation. No Vietnamese benchmark yet addresses *multimodal* misinformation (image + text claims) a critical gap given the prevalence of image-based misinformation on Vietnamese social platforms.

3. Modeling approaches: From neural representations to foundation models

The trajectory of Vietnamese NLP reflects a profound paradigm shift from rigid statistical frameworks to massively parallel neural architectures. This section critically analyzes the causal relationships driving this evolution rather than treating individual models as isolated contributions.

3.1. Neural language models (2016–2020)

The first deep learning era for Vietnamese NLP was defined by Recurrent Neural Networks (RNNs) (Tran et al., 2016) and Long Short-Term Memory (LSTM) networks (Nguyen et al., 2018e). Unlike prior statistical approaches, LSTMs introduced gated memory mechanisms—input, forget, and output gates—capable of regulating information flow across variable-length sequences. In 2016, Tran et al. (2016) proposed character-level and syllable-level RNN architectures enhanced with LSTM cells to model Vietnamese linguistic features such as syllable composition and diacritic marks, achieving a perplexity of 83.8 on a conversational corpus of 24 million syllables derived from 1500 movie

Table 17

Publicly available datasets for 16 tasks in the Vietnamese language.

Dataset	Size	Task	Domain / Source
Dependency Parsing			
VnDT (Nguyen et al., 2014b)	10K	Dependency Parsing	General treebank
VLSP2020-UD (Linh et al., 2020)	9K	Dependency Parsing	Universal Dependencies
Semantic Parsing			
ViText2SQL (Nguyen et al., 2020a)	10K	Semantic Parsing	Structured SQL queries
VLSP2025-viSemParse (Linh et al., 2025)	2.5K	Semantic Parsing	AMR graph annotation
Named Entity Recognition			
EVNECorpus (Ngo et al., 2014)	45K	NER	Bilingual aligned news
VLSP2016-NER (Nguyen et al., 2019a)	16K	NER	Vietnamese news articles
VLSP2018-NER (Nguyen et al., 2019a)	1282 articles	NER	VnExpress, Zingnews
VLSP2021-NER (Linh et al., 2022)	2106 articles	NER	VnExpress, Baomoi, Zingnews
PhoNER_COVID19 (Truong et al., 2021)	10K	NER	Healthcare / COVID-19
POS Tagging			
VLSP2013 – POS Tagged Corpus (Nguyen et al., 2009)	27K	POS Tagging	Online news articles
NIIVTB (Nguyen et al., 2018d)	20K	POS Tagging	Socio-political news
Word Segmentation			
VLSP2013 – Word-Segmented Corpus (Nguyen et al., 2009)	77K	Word Segmentation	General news corpora
NIIVTB (Nguyen et al., 2018d)	20K	Word Segmentation	Socio-political news
Relation Extraction			
VLSP2020-RelEx (Mai et al., 2020)	1056 docs	Relation Extraction	News (LOC / ORG / PER)
Semantic Textual Similarity			
ViCon (Nguyen et al., 2018a)	1400 pairs	Semantic Similarity	Synonym / antonym pairs
ViSim-400 (Nguyen et al., 2018a)	400 pairs	Semantic Relatedness	Five semantic relation types
Natural Language Inference (NLI)			
ViNLI (Huynh et al., 2022)	30K	NLI	Open-domain news
vnNLI – VLSP 2021 (Quyen et al., 2022)	20K	NLI	Healthcare news
VnNewsNLI (Quyen et al., 2022)	44K	NLI	VnExpress (native)
ViANLI (Van Huynh et al., 2026)	10K	NLI	Open-domain news
Machine Translation			
PhoMT (Doan et al., 2021)	3M	Machine Translation	Wikipedia, TED Talks
MedEV (Vo et al., 2024)	360K	Machine Translation	Medical journals, clinical manuals
ViBidirectionMT – Eval (Viet et al., 2025)	100K	Machine Translation	Bidirectional VI↔EN
QA and Machine Reading Comprehension (MRC)			
UIT-ViQuAD (Nguyen et al., 2020b)	23K Q&A	Extractive MRC	Vietnamese Wikipedia
UIT-ViQuAD 2.0 (Kiet et al., 2022)	36K Q&A	Extractive MRC	Vietnamese Wikipedia
UIT-ViNewsQA (Van Nguyen et al., 2022)	22K Q&A	MRC	Vietnamese news
ViHealthQA (Nguyen et al., 2022c)	10K Q&A	MRC	Medical / health news
ViCoQA (Luu et al., 2021a)	10K Q&A	Conversational MRC	Open-domain
ViMMRC (Nguyen et al., 2020c)	2.7K Q&A	Multiple-choice MRC	School literature
ViMMRC 2.0 (Luu et al., 2025)	23K Q&A + 5K passages	Multiple-choice MRC	Extended school curricula
VIMQA (Le et al., 2022)	10K Q&A	Multi-hop QA	Vietnamese Wikipedia
Open-ViTabQA (Dao et al., 2025)	9.9K Q&A	Tabular QA	Structured tables
ViVQA (Tran et al., 2021a)	15K Q&A & 10K imgs	Visual QA	MS COCO adapted
VLSP2022-EVJVQA (Luu-Thuy Nguyen et al., 2023)	33K Q&A & 5K imgs	Visual QA	Multilingual visual
OpenViVQA (Nguyen et al., 2023f)	37K+ Q&A & 11K+ imgs	Visual QA	Native Vietnamese imagery
ViCLEVR (Tran et al., 2024)	30K+ Q&A & 25K+ imgs	Visual QA	Synthetic visual reasoning
ViTextVQA (Van Nguyen et al., 2025)	50K Q&A & 16K imgs	Visual QA	OCR / scene text
UIT-ViCoV19QA (Thai et al., 2022)	4.5K Q&A	Community QA	COVID-19 discussion forums
COVID-KGQA (Tran et al., 2022a)	1K Q&A	Knowledge-based QA	COVID-19 knowledge graphs
VlogQA (Ngo et al., 2024)	10K Q&A	Spoken MRC	Video transcripts
VLSP2025-FinQA (Toan et al., 2025)	4K	Numerical Q&A	Financial reports

subtitles—demonstrating that RNN-based models substantially outperform bi-gram baselines on Vietnamese.

Bidirectional LSTMs (Bi-LSTMs) (Nguyen et al., 2018c) advanced this further by enabling simultaneous access to both preceding and following context, proving highly effective for NER and sentiment analysis. Nguyen et al. (2018e) introduced the hybrid LD-SVM approach, integrating LSTM hidden state representations with a Support Vector Machine classifier; the combined model achieved a 90.2% F1-score on UIT-VSFC (Nguyen et al., 2018b), with enhanced efficacy on longer syn-

tactic constructions. Independently, Nguyen et al. (2018c) demonstrated that the Bi-LSTM architecture attained 92.0% F1 for ternary sentiment classification and 89.6% F1 for multi-class topic categorization on the same corpus (Nguyen et al., 2018b), establishing Bi-LSTM as the robust baseline for Vietnamese contextual sequence labeling in low-resource settings.

The fundamental limitation of this era was computational: sequential processing prevented parallelization across the token dimension, making large-corpus training prohibitively slow. More critically, LSTMs

Table 17
Continued.

Dataset	Size	Task	Domain / Source
Sentiment Analysis (SA)			
UIT-VSFC (Nguyen et al., 2018b)	16K	Traditional SA	University student feedback
VLSP2016-SA (Nguyen et al., 2019b)	6.4K	Traditional SA	Tech review sites, Facebook
VLSP2018-ASA (Nguyen et al., 2019b)	11K	Aspect-based SA	Restaurant and hotel reviews
UIT-VSMEC (Ho et al., 2020)	7K	Emotion Recognition	Social media (6 emotion labels)
UIT-ViSFD (Luc Phan et al., 2021)	11K	Aspect-based SA	Mobile e-commerce (ABSA)
UIT-ViOCD (Nguyen et al., 2021c)	5.4K	Traditional SA	E-commerce, Google Play
ViCTSD (Nguyen et al., 2021a)	10K	Toxic Speech	Social media comments
UIT-ViHSD (Luu et al., 2021b)	33K	Hate Speech	Facebook comments
UIT-ViHOS (Hoang et al., 2023)	26K	Hate Speech	Annotated social media spans
VLSP 2019 HSD (Vu et al., 2020)	25K	Hate Speech	Facebook textual items
Summarization			
VNDS (Nguyen et al., 2019c)	>150K	Summarization	Multi-document news
VSolSCSum (Nguyen et al., 2016)	3.7K	Summarization	Single-document news
ViMs (Tran et al., 2020)	300 clusters	Summarization	Multi-document news
Text Classification			
UIT-ViNames (To et al., 2021)	26K	Gender Classification	Student name lists
UIT-ViON (Tran et al., 2021b)	260K	News Classification	Online newspapers (multi-label)
Legal Engineering			
ALQAC series (Do et al., 2025a, 2024a; Nguyen et al., 2022a, 2023a; Thanh et al., 2021b)	~1,000	Legal Retrieval & QA	Statutory and civil law
R2GQA (Do et al., 2025b)	9758	Legal Retrieval & QA	Regulation-to-graph
ViBidLQA (Ha et al., 2024)	3013	Legal Retrieval & QA	Bidirectional legal QA
Vietnamese Labor Law QA (Pham et al., 2024)	19,330	Legal Retrieval & QA	Labor law domain
VLQA (Nguyen et al., 2025)	3K	Legal QA	General legal texts
VLSP2025-MLQA-TSR (Son et al., 2025)	~800	Legal Retrieval & QA	Multimodal legal documents
VLSP2025-DRILL (Vuong et al., 2025)	2K	Legal Retrieval	Legal document retrieval
Fact-Checking			
ReINTEL (Le et al., 2020)	10K	Fake News Detection	Internet claim verification
ViWikiFC (Le et al., 2024)	20K	Claim Verification	Wikipedia evidence
ViFactCheck (Hoa et al., 2025)	7232	Multi-hop Verification	Multi-source evidence
ViNumFCR (Luong et al., 2025)	10K	Numerical Verification	Numerical fact claims

Table 18

Comparative evaluation of NLP architectural paradigms for Vietnamese NLP.

Architectural Paradigm	Primary Mechanism	Advantages for Vietnamese NLP	Inherent Limitations
Sequential Neural Networks (RNN, Bi-LSTM)	Sequential processing with gated memory cells for local context retention.	Captures local syntactic dependencies; robust baseline for sentiment analysis and sequence labeling tasks.	Sequential bottleneck prevents parallelization; limited context window fails to resolve long-distance anaphora in pro-drop Vietnamese sentences.
Multilingual Transformers (XLM-R, mBERT)	Parallel self-attention over shared cross-lingual subword vocabularies trained across 100+ languages.	Enables zero-shot cross-lingual transfer from high-resource languages; robust on generalized comprehension tasks.	Generalized tokenizers inappropriately fragment Vietnamese syllables; diluted language-specific depth due to the “curse of multilinguality.”
Monolingual Transformers (PhoBERT, ViDeBERTa, BARTpho, CafeBERT)	Parallel self-attention with language-specific byte-level BPE tokenization pre-trained on Vietnamese corpora.	Unparalleled accuracy on core Vietnamese NLP tasks (POS, NER, NLI); captures exact orthographic, syntactic, and tonal nuances of Vietnamese.	Requires massive, high-quality monolingual corpora for pre-training; encoder-only variants limit direct generative applications.
Large Language Models (PhoGPT, Vistral, SeaLLM)	Decoder-only autoregressive generation scaled to billions of parameters with instruction alignment.	Capable of complex reasoning, open-ended generation, zero-shot instruction following, and conversational fluency in Vietnamese.	Computationally prohibitive for local deployment; prone to intrinsic hallucinations; highly sensitive to prompt phrasing and regional dialect variations.

struggled to maintain reliable attention across long Vietnamese sentences where key semantic dependencies span 20+ tokens—particularly in the multi-clause constructions and pro-drop structures common in formal and legal Vietnamese text. These limitations directly motivated the transition to Transformer-based architectures described in Section 3.2.

3.2. Transformer-based language models

The Transformer architecture (Vaswani et al., 2017) eliminated the sequential processing bottleneck of LSTMs entirely by replacing recurrence with parallel self-attention, enabling the modeling of arbitrary-length dependencies across the token dimension and drastically accelerating training on large corpora. For Vietnamese NLP, this trig-

gered a critical methodological divergence across four functional categories: (1) cross-lingual baseline systems, (2) multi-task linguistic toolkits, (3) monolingual pre-trained encoder and encoder-decoder models, and (4) retrieval-augmented QA systems. Each category addresses a distinct gap in the Vietnamese NLP pipeline, and their collective development defines the current state of the art.

Cross-Lingual Baselines. The first wave of Vietnamese Transformer applications leveraged existing multilingual models as off-the-shelf backbones. **XLMRQA** (Nguyen et al., 2022b) was the first Vietnamese QA system utilizing a Transformer backbone, fine-tuning XLM-RoBERTa-Large (Conneau et al., 2020) on the UIT-ViQuAD corpus—a Vietnamese Wikipedia-derived dataset comprising 23,074 question-answer pairs

across 5109 passages—within a retriever–reader–selector framework integrating the Pyserini retriever (Lin et al., 2021), achieving 64.99% F1 and 51.94% EM. The system identified complex *Why* and *How* queries requiring multi-sentence inference as the primary failure mode, emphasizing the necessity of adapting multilingual models to address long-context comprehension challenges in Vietnamese. **ViQAS** (Van Nguyen et al., 2023) extended this direction with a multi-stage transfer learning pipeline built on BERTology-family models, specifically designed to integrate Vietnamese-specific question word diversity and Transformer-based evidence extraction, directly addressing the limited ability of standard BERT variants to encode longer texts containing multiple distractor sentences. Both systems empirically confirmed that generalized multilingual tokenizers fragment Vietnamese syllables, motivating the development of language-specific alternatives.

Multi-Task Linguistic Toolkits. Two widely adopted toolkits unified core Vietnamese NLP pipelines within a single Transformer framework. **PhoNLP** (Nguyen & Nguyen, 2021), licensed under Apache 2.0, integrates a shared PhoBERT-based (Nguyen & Tuan Nguyen, 2020) encoding layer with three independent decoding heads for POS tagging, NER, and dependency parsing, enabling simultaneous multi-task annotation in a single forward pass. **Trankit** (Nguyen et al., 2021b) adopts a complementary multilingual approach, leveraging XLM-RoBERTa-Large (Conneau et al., 2020) to provide trainable pipelines for seven NLP tasks—sentence segmentation, tokenization, multi-word token expansion, POS tagging, morphological feature tagging, dependency parsing, and NER—across 100 languages, with 90 downloadable pre-trained pipelines for 56 languages including Vietnamese.

Monolingual Pre-Trained Encoder and Encoder-Decoder Models. Recognizing the systematic limitations of multilingual tokenizers, the community developed a progression of Vietnamese-specific pre-trained models. **PhoBERT** (Nguyen & Tuan Nguyen, 2020), developed by VinAI Research, is the first large-scale monolingual Vietnamese model based on the RoBERTa architecture (Zhuang et al., 2021), pre-trained on a 20GB Vietnamese corpus with a custom byte-level BPE tokenizer, establishing new state-of-the-art results on POS tagging, NER, and dependency parsing. **ViHealthBERT** (Minh et al., 2022) extended this paradigm to the medical domain via secondary domain-adaptive pre-training that leverages PhoBERT's pre-trained weights, achieving state-of-the-art on three downstream tasks: COVID-19 NER, acronym disambiguation, and healthcare summarization. **ViSoBERT** (Nguyen et al., 2023c) addressed the informal social media domain, constructed on an XLM-R-style architecture (768 hidden units, 12 self-attention layers, 12 attention heads) and pre-trained using the Masked Language Model objective on 1GB of uncompressed text crawled from Facebook, TikTok, and YouTube; its custom SentencePiece tokenizer is specifically engineered to handle Vietnamese “teencode,” emojis, and diacritic-omitted text, achieving superior coverage and semantic representation over prior models on hate speech and social sentiment tasks. **ViDeBERTa** (Tran et al., 2023b) is based on the DeBERTaV3 (He et al., 2021) architecture and pre-trained on 138GB of Vietnamese text from the CC100 corpus (Conneau et al., 2020)—nearly 7× larger than PhoBERT's pre-training data—with disentangled attention and gradient-disentangled embedding sharing across three variants (xsmall: 22M, base: 86M, large: 304M parameters); ViDeBERTa achieves 97.2% accuracy on POS tagging (vs. PhoBERT's 96.8%), 95.3% F1 on NER (vs. ViT5's 94.5%), and 89.9% F1 on MRC (vs. XLM-R's 87%), despite ViDeBERTa-base using only 23% of PhoBERT-large's parameters—empirically confirming the data-scaling hypothesis over parameter-count scaling. **CafeBERT** (Do et al., 2024b), initialized from XLM-RoBERTa-Large (Conneau et al., 2020) and subjected to secondary pre-training via the Masked Language Model objective on 18GB of Vietnamese text (180M sentences; 2.8B word tokens sourced from Vietnamese Wikipedia and de-duplicated online news), achieves 92.47% F1 on UIT-ViQuAD 2.0—a 17-point gap over mBERT (75.33% F1)—and currently represents the state-of-the-art across all five

VLUE benchmark tasks, confirming that language-specific secondary pre-training is the dominant performance driver over multilingual coverage (Table 18). **BARTpho** (Tran et al., 2022b) is the first publicly available large-scale Vietnamese seq2seq model, released as BARTpho-syllable and BARTpho-word based on the BART denoising autoencoder architecture (Lewis et al., 2020); it achieves a ROUGE-2 score of 38.4 on VNDS summarization (+5.1 over PhoBERT-based extractive baselines) and up to +3.2 BLEU on PhoMT machine translation over standard Transformer baselines. **ViT5** (Phan et al., 2022) adopts a T5-style encoder-decoder architecture pre-trained on the CC100 Vietnamese corpus with self-supervised span-corruption objectives, benchmarking competitively on abstractive summarization and NER, and serving as the backbone for the ViText2SQL semantic parsing dataset.

Retrieval-Augmented QA Systems. Two dedicated systems advanced Vietnamese open-domain and legal QA beyond standard fine-tuning. **ViWiQA** (Nguyen et al., 2023c) is the first Vietnamese open-domain QA system operating over the full Vietnamese Wikipedia, employing XLM-RoBERTa (Conneau et al., 2020) as its core backbone for both the span-extraction Reader and the passage-relevance Cross-Encoder; its retriever improves top-1 passage retrieval accuracy by 20% over BM25, yielding 15-point and 17-point absolute gains in Exact Match and F1 on single-hop QA, and a 4-point gain in correct passage-pair retrieval on multi-hop QA. **Vuong et al. (2023)** introduced an end-to-end article-level retrieval-based Vietnamese legal QA system that addresses data scarcity through automatic weak-label generation from legal article titles and contents; their pipeline combines a BM25-based quickview retriever with a BERT classifier fine-tuned sequentially on weak then gold labels, achieving an F_2 score of 0.6007 on a corpus of 117,557 civil law articles—22% higher than either component alone—while maintaining real-time inference at approximately 1–3 s per query.

3.3. Multimodal learning models

The multimodal frontier of Vietnamese NLP integrates visual and textual modalities to address tasks such as Visual Question Answering (VQA), scene-text comprehension, and layout-aware document intelligence. The architectural progression mirrors the broader NLP trajectory: from adapted co-attention baselines through encoder–decoder fusion architectures to layout-aware Transformers capable of reasoning over native Vietnamese documents.

Cross-Modal Co-Attention Baselines. The first wave of Vietnamese multimodal models adapted state-of-the-art English VQA architectures to the Vietnamese context using the ViVQA dataset (Tran et al., 2021a) as the primary benchmark. **ViCAN** (Phuc et al., 2021) is a revised version of mCAN (Yu et al., 2019), the championship model of the VQA Challenge 2019 held in conjunction with the CVPR conference. mCAN comprises cascaded Modular Co-Attention (MCA) layers, with each layer modeling self-attention for questions and images, as well as question-guided attention for images using a modular composition of two basic attention units. ViCAN converts the image and question into a visual feature matrix and contextual feature matrix respectively, applies co-attention layers, combines the outputs via a fusion layer to obtain a joint representation, and predicts answers through a non-linear layer followed by linear mapping. **BiCAN** (Nguyen-Tran et al., 2022) extended this with a novel Bi-directional Cross-Attention Network tailored for Vietnamese VQA. The method integrates a Vision Transformer (ViT) to extract patch-based visual embeddings via [CLS] token aggregation and PhoBERT to process textual questions. The Bi-directional Cross-Attention module facilitates simultaneous mutual learning between visual and textual representations via distinct multi-head attention layers, additive feature combination, and multiplicative fusion followed by softmax classification, achieving state-of-the-art **51.3%** accuracy on ViVQA, significantly outperforming LSTM (33.85%), Bi-LSTM (33.97%), and Hierarchical Co-Attention (34.96%) baselines. **The Hierarchical**

Co-Attention model (Tran et al., 2021a) employs co-attention to utilize feature information from both an image and a question in a logical manner, recognizing that image features attentively support question features and vice versa. Phrase-level features obtained through max-pooling are encoded via LSTM, and an Alternating Co-Attention mechanism alternately attends to question or image features guided by the complementary modality.

Multilingual and Sequence-to-Sequence VQA. Triet (Thai & Luu, 2024) addressed the multilingual VLSP2022-EVJVQA task by extracting visual features and hints from image-question pairs using a Vision Transformer (ViT) and cutting-edge vision-and-language models such as ViLT and OFA. The visual features and hints were then combined with questions and fed into a ConvS2S (Convolutional Sequence-to-Sequence) network to generate answers. Because the CNN-based backbone enables full parallelization during training, the model effectively utilizes GPU hardware, achieving a competitive third-ranked result on the open-ended and free-form-answer VLSP2022-EVJVQA task. **BARTPhoBEiT** (Tran et al., 2023c) is an innovative fusion of BARTpho (Tran et al., 2022b) and BEiT-3 (Wang et al., 2023), specifically designed for the Vietnamese language. This model incorporates pre-trained Sequence-to-Sequence and bidirectional encoder representations derived from Image Transformers, and is extensively evaluated on Vietnamese VQA datasets, demonstrating its effectiveness and suitability for open-ended answer generation in Vietnamese. **OpenViVQA** (Nguyen et al., 2023f) introduced both a large-scale native Vietnamese VQA dataset of 37K question-answer pairs across 11K images and a family of multimodal fusion models specifically designed for open-ended answer generation over authentically Vietnamese scenes, establishing the first benchmark that moves beyond COCO-adapted imagery to native Vietnamese visual environments.

Transformer-CNN Fusion and Visual Reasoning. **AViVQA-TranConI** (Nguyen et al., 2024) introduced a novel framework combining Transformer-based global feature extraction via BLIP-2 and CNN-based local feature extraction via EfficientNet-B7 to obtain complementary visual representations. For text processing, BARTpho word encodes questions, and a multimodal fusion module based on BEiT-3 (Wang et al., 2023) dynamically aligns visual and textual embeddings through a mixture-of-experts architecture. The fused features are processed by a classifier with adaptive pooling and normalization layers, achieving 71.04% accuracy on ViVQA, outperforming BARTPhoBEiT (68.58%) and MCA (60.76%). ViCLEVR (Tran et al., 2024) simultaneously introduced a synthetic visual reasoning dataset of 30K question-answer pairs across 25K images and a hybrid multimodal fusion model. Unlike datasets derived from real-world images, ViCLEVR generates structured compositional scenes requiring multi-step spatial and relational reasoning, thereby probing systematic generalisation rather than pattern matching. The accompanying hybrid fusion model integrates a CNN-based visual feature extractor with a Vietnamese pre-trained language encoder, demonstrating that compositional reasoning benchmarks expose brittleness in standard attention-based fusion approaches not revealed by standard VQA datasets.

Scene-Text VQA and Layout-Aware Document Intelligence. **ViTextVQA** (Van Nguyen et al., 2025) introduced a large-scale scene-text VQA dataset of 50K question-answer pairs across 16K images containing Vietnamese text in natural scenes, alongside a novel multimodal feature fusion method. Unlike standard VQA where answers derive from visual objects, ViTextVQA requires models to jointly perform OCR-level text recognition and natural language understanding; the proposed fusion method integrates visual features from a ViT encoder with OCR-extracted token embeddings via cross-modal attention, substantially outperforming generic multilingual scene-text models that lack Vietnamese-specific tokenization. **LiGT** (Le et al., 2025) is the most architecturally sophisticated Vietnamese multimodal model to date,

Table 19

Model performance on Vietnamese visual question answering benchmarks (Accuracy, %). Best results are **bolded**.

Model	Backbone	ViVQA
LSTM baseline	LSTM + CNN	33.85
Bi-LSTM baseline	Bi-LSTM + CNN	33.97
Hier. Co-Attention	LSTM + Co-Attention	34.96
ViCAN	mCAN + PhoBERT	48.10
BiCAN	ViT + PhoBERT	51.30
BARTPhoBEiT	BARTpho + BEiT-3	68.58
AViVQA-TranConI	BLIP-2 + EfficientNet-B7	71.04

designed for layout-aware document understanding tasks including invoice processing, legal certificate parsing, and government form comprehension. LiGT incorporates *LayoutHEI* (Layout Hashing-based Embedding Infusion), which converts OCR text coordinates into hierarchical spatial embeddings through recursive image quartering across four levels, mapped to embeddings using existing model layers to maintain parameter efficiency without introducing new neural modules. LiGT's multimodal fusion combines OCR text, visual features extracted via a convolutional backbone, and layout embeddings within a ViT5-based encoder-decoder framework, dynamically weighting layout and text embeddings through a learnable ratio ($\omega = 0.55$) to enhance adaptability. LiGT demonstrates superior performance over extractive baselines such as LayoutLM and BARTPhoBEiT by 5.2–37.2% in accuracy, particularly excelling with long answers and ambiguous queries, while ablation studies confirm the necessity of layout hashing (optimal at four levels) and the learnable ratio. Evaluations on English benchmarks (DocVQA, InfographicVQA) using Flan-T5 achieve a competitive 63.8% ANLS, highlighting LiGT's cross-lingual capabilities.

Duy-Minh. Beyond the architectural design described above, we further analyze the empirical behavior of Nguyen-Tran et al. (2022) from an evaluation perspective. In 2022, Nguyen-Tran et al. (2022) propose a Vietnamese-specific Visual Question Answering model that integrates a Vision Transformer for image encoding and PhoBERT for text encoding within a novel Bi-directional Cross-Attention framework, enabling simultaneous mutual refinement of visual and linguistic representations. Evaluated on the ViVQA benchmark—comprising 15,000 Vietnamese-adapted MS COCO samples—the model attains **51.3%** accuracy, markedly surpassing existing baselines such as Hierarchical Co-Attention (34.96%), LSTM (33.85%), and Bi-LSTM (33.97%). Ablation studies confirm the efficacy of the bi-directional attention mechanism and multiplication-based feature fusion, while the model's simplicity—employing a single attention block—reflects dataset size constraints. **Table 19** summarises the empirical progression of Vietnamese multimodal VQA models, demonstrating a consistent accuracy improvement from co-attention baselines to Transformer-CNN fusion architectures.

3.4. Large language models and cross-lingual architectures

The contemporary paradigm has decisively pivoted toward billion-parameter generative foundation models. While global LLMs such as GPT-4 achieve reasonable Vietnamese performance via cross-lingual transfer, their English-centric tokenizers and lack of cultural alignment limit their effectiveness. The Vietnamese research community has responded with a growing ecosystem of dedicated foundation models, outlined below and benchmarked in **Table 20**.

3.4.1. Cross-lingual transfer strategies

For low-resource languages, cross-lingual transfer learning constitutes a critical methodology, leveraging abundant annotated data and pre-trained weights from high-resource languages to bootstrap Vietnamese NLP systems (Conneau et al., 2020; Ruder & Korashy, 2019). The

foundational mechanism relies on shared subword representations and aligned semantic spaces, enabling concepts to be learned independently of the specific input language. Experimental analyses indicate that zero-shot cross-lingual transfer is highly effective for semantic textual similarity (Nguyen et al., 2018a) and basic sentiment analysis (Nguyen et al., 2018b), but degrades significantly on complex tasks such as machine reading comprehension (Nguyen et al., 2020b) or natural language inference (Huynh et al., 2022), where language-specific syntactic nuances are paramount. Advanced strategies to enhance transfer efficacy include: (i) data-based transfer via machine-translated training corpora (Doan et al., 2021); (ii) constrained decoding using linguistic priors; (iii) phonetic transliteration to capture cognate relationships; and (iv) multilingual colexification graphs that link concepts across typologically diverse languages without relying on parallel corpora.

3.4.2. Vietnamese-developed LLMs

The first major indigenous effort is **PhoGPT** (Nguyen et al., 2023d), released by VinAI Research. The base model PhoGPT-4B contains exactly 3.7B parameters, pre-trained from scratch on a Vietnamese corpus of 102 billion tokens with an 8192-token context length and a custom 20,480-type byte-level BPE vocabulary. Its chat variant, PhoGPT-4B-Chat, is further fine-tuned on 290K Vietnamese conversations. This from-scratch training strategy gives PhoGPT superior tokenization fidelity compared to models adapted from English-centric bases.

VinaLLaMA (Nguyen et al., 2023g) takes a complementary approach by building on LLaMA-2 with an additional 800 billion training tokens specifically curated for Vietnamese. Notably, VinaLLaMA was designed not only for linguistic fluency but also for cultural alignment it embeds knowledge of Vietnamese customs, history, and social context, making it the first truly indigenised Vietnamese LLM. VinaLLaMA-7B-Chat, fine-tuned on 1 million high-quality synthetic samples, achieves state-of-the-art results across the VLSP, VMLU, and Vicuna Benchmark Vietnamese evaluations.

Vistral (Nguyen et al., 2023b) is built upon Mistral-7B-v0.1, using grouped-query attention and sliding window attention, and applies an additional phase of continual pre-training specifically adapted to Vietnamese. It achieves 50.07% average VMLU accuracy, significantly surpassing ChatGPT (46.33%) despite identical parameter count, demonstrating that language-specific alignment outweighs raw scale for Vietnamese-specific reasoning. **Vi-Mistral-X** (Vo, 2024) extends this Mistral-based continual pre-training approach, outperforming existing Vietnamese LLMs on text classification, question answering, and generation tasks and setting a new VMLU standard at the time of its release.

Further expanding the ecosystem, **Truong et al. (2024)** released a suite of fine-tuned open-source Vietnamese LLMs: **URA-LLaMA** (7B, 13B, 70B parameters, based on LLaMA-2), **GemSura** (2B and 7B, based on Gemma), and **MixSura** (8×7B, based on Mixtral). This work was the first to release Vietnamese LLMs at the 13B and 70B scales, demonstrating that fine-tuning with LoRA, quantization, and large batch sizes effectively transfers cross-language knowledge to Vietnamese. Industry has also made significant contributions: **KiLM** (Zalo AI), at 13B parameters, has reached the top-5 on the VMLU leaderboard, directly competing with Meta's Llama-3-70B and Google's Gemini.

3.4.3. Regional foundation models

Beyond purely Vietnamese models, regional multilingual pre-training has produced strong results. **SeaLLM** (LLaMA-2/Mistral, 7B) and **Sailor** (Qwen 1.5, 0.5B–14B, pre-trained on 200–400B tokens across 7 SEA languages including Vietnamese, Thai, Indonesian, Malay, and Lao) demonstrate that regional multilingual pre-training yields competitive Vietnamese commonsense reasoning while maintaining cost-effective tokenization for diacritic-heavy scripts. These regional models are particularly suited to cross-lingual applications requiring joint Vietnamese–English or Vietnamese–Thai processing.

3.4.4. VMLU benchmark evaluation

The rigorous evaluation of Vietnamese NLP has necessarily evolved from reporting isolated metrics to deploying unified, multi-domain benchmark suites.

The VLUE Benchmark (Do et al., 2024b) standardized comparative model evaluation by providing a unified suite of five NLU tasks text classification, span extraction, NLI (Huynh et al., 2022), POS tagging (Nguyen et al., 2009), and hate speech detection (Luu et al., 2021b). Empirical results from VLUE consistently confirm the supremacy of customized, domain-adapted architectures over generic multilingual models, with **ViDeBERTa** (Tran et al., 2023a) achieving consistent state-of-the-art across all five evaluated tasks.

The VMLU Benchmark (Bui et al., 2025) was developed to evaluate the complex reasoning and instruction-following capabilities of LLMs in Vietnamese, comprising multiple-choice questions across STEM, Humanities, Social Sciences, and Professional domains strictly adapted to Vietnamese cultural and educational standards.³ As shown in Table 20, global models benefit substantially from Southeast Asian training data inclusion: Qwen2.5-72B-Instruct (69.2%) overtakes GPT-4 (65.5%) in zero-shot performance, confirming that targeted regional pre-training outweighs raw parameter scale. Among Vietnamese-developed models, **Vistral-7B-Chat** (Nguyen et al., 2023b) achieves 50.07% surpassing ChatGPT and demonstrating the effectiveness of language-specific continual pre-training while larger open-weight models such as QwQ-32B dominate the leaderboard overall, underscoring that cost-effective, locally deployable solutions remain a critical open challenge for Vietnamese enterprise adoption.

Table 20 demonstrates that regional multilingual pre-training (Qwen2.5-72B-Instruct, 69.17%) outperforms English-centric global models (GPT-4, 65.53%) in zero-shot Vietnamese evaluation, confirming that Southeast Asian training data provides measurable benefits over raw parameter scale. Among Vietnamese-developed models, Vistral-7B-Chat (50.07%) surpasses ChatGPT (46.33%) despite identical parameter count, demonstrating the effectiveness of language-specific continual pre-training as a cost-efficient alternative to billion-parameter scaling.

3.5. Evaluation metrics

Recall, Precision and F1. In Vietnamese NLP, recall, precision, and F1 score are commonly used evaluation metrics for tasks such as named entity recognition and text classification. Precision measures how accurate the positive predictions of a model or system are. It is calculated as the ratio of true positives (correctly predicted positive instances) to the sum of true positives and false positives (instances predicted as positive but actually negative).

Recall measures how well a model or system is able to identify all positive instances in a dataset. It is calculated as the ratio of true positives to the sum of true positives and false negatives (instances that were not predicted as positive but were actually positive). The F1 score, which is the harmonic mean of precision and recall, is also commonly used to balance the tradeoff between these two metrics. They are calculated as follows:

$$Recall = \frac{\sum_{i=1}^n \sum_{j=1}^n a_{ij}}{\sum_{i=1}^n \sum_{j=1}^n \hat{y}_i} \quad (20)$$

$$Precision = \frac{\sum_{i=1}^n \sum_{j=1}^n a_{ij}}{\sum_{i=1}^n \sum_{j=1}^n \hat{y}_j} \quad (21)$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \quad (22)$$

where n is the number of data points, a_{ij} is the number of correct predictions for the i^{th} class and j^{th} sample, y_i is the total number of actual samples of the i^{th} class, and $\hat{y}_j$ is the total number of predicted samples of the j^{th} class.

³ <https://vmlu.ai/leaderboard>

Table 20
Model performance on VMLU benchmark (Zero-Shot) (Bui et al., 2025).

Category	Model	Params	STEM	Social	Hum.	Avg
Global Models	QwQ-32B (Alibaba)	32B	81.11	78.49	71.78	76.13
	Qwen2.5-72B-Instruct (Alibaba)	72B	68.46	74.65	69.23	69.17
	Llama-3-70B (Meta)	70B	61.70	74.91	68.74	66.44
	GPT-4 (OpenAI)	N/D	63.84	71.78	66.14	65.53
	ChatGPT (OpenAI)	N/D	43.24	51.67	46.96	46.33
Vietnamese-Developed Models	MISA-AI-1.0 (MISA JSC)	N/D	84.34	83.58	79.05	81.26
	Vi-Sovereign-Medium (NLP-CORE-Lab)	N/D	86.78	83.20	75.55	80.57
	VNPTAI.IO-Medium-R1.2 (VNPT AI)	N/D	84.36	82.22	76.42	79.61
	GreenNode-xMedium-v1	N/D	75.70	81.09	75.25	75.50
	KiLM-13b-v24.7.1 (Kiki AI)	13B	60.29	73.15	71.85	66.07
	Vistral-7B-Chat	7B	43.32	57.02	55.12	50.07
	PhoGPT-7B5-Instruct (VinAI)	7.5B	21.97	25.93	24.32	24.01
	ura-llama-7b (HCMUT)	7B	29.19	33.31	34.64	32.18
	vinallama-7b-chat	7B	25.70	34.50	33.87	30.64

Table 21
A summarization of Vietnamese NLP evaluation metrics.

Metric	Formula/Description	Usage
Precision	$Precision = \frac{\sum_{i=1}^n \sum_{j=1}^n a_{ij}}{\sum_{i=1}^n \sum_{j=1}^n \hat{y}_{ij}}$	Measures accuracy of positive predictions.
Recall	$Recall = \frac{\sum_{i=1}^n \sum_{j=1}^n a_{ij}}{\sum_{i=1}^n \sum_{j=1}^n y_{ij}}$	Measures ability to identify all positive instances.
F1-Score	$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$	Harmonic mean of precision and recall; strikes a balance between the two.
Accuracy	$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$	Evaluates predictive performance of the model.
Accuracy@k	$Acc@k = \frac{1}{n} \sum_{i=1}^n \frac{1}{k} \sum_{j=1}^k \delta(y_{ij}, \hat{y}_{ij})$	Measures accuracy of the top-k predictions.
BLEU	$BLEU = \exp \left(\min \left(0, 1 - \frac{\sum \dots}{\sum \dots} \right) \right)$	Evaluates quality of machine translation output.
Exact Match	$EM = \frac{1}{N} \sum_{i=1}^N \delta(y_i, \hat{y}_i)$	Evaluates performance on QA & MRC tasks.
ROUGE-N	$ROUGE-N = \frac{\sum_{g \in m} \min(\dots)}{\sum_{g \in c} \text{count}_c(g)}$	Measures n-gram overlap between generated and reference summaries.
ROUGE-L	$ROUGE-L = \frac{\sum_{s \in S} \text{LCS}(s, g(s))}{\sum_{s \in S} \text{len}(s)}$	Measures longest common subsequence overlap.
ROUGE-S	$ROUGE-S = \frac{\sum_{(a,b) \in A} \text{LCS}(a, b)}{\sum_{(a,b) \in A} b }$	Measures skip-bigram overlap.
ROUGE-W	$ROUGE-W = \frac{\sum_{i=1}^n W_i}{\sum_{i=1}^n C_i}$	Measures weighted n-gram overlap.
METEOR	$METEOR = F_{mean} \cdot (1 - Penalty)$	Harmonic mean of unigram precision and recall (recall weighted higher).
CIDEr	$CIDEr(c_i, S_i) = \sum_{n=1}^N \frac{w_n}{m} \sum_j \dots$	Evaluates image captioning; captures human consensus.
SPICE	$SPICE(c, S) = \frac{1}{\frac{1}{P(c, S)} + \frac{1}{R(c, S)}}$	Measures quality of image captions via scene graphs.
BERTScore	$BERTScore = \frac{1}{\frac{1}{P_{BERT}} + \frac{1}{R_{BERT}}}$	Assesses quality of text generation via embeddings.

Accuracy. Accuracy is a critical evaluation metric in various NLP tasks, including sentiment analysis, named entity recognition, machine translation, and text classification. High accuracy scores demonstrate the model's ability to precisely process and comprehend Vietnamese language data. Conversely, low accuracy scores suggest the need for further training or system tuning to enhance its performance.

$$Accuracy = \frac{\#Correct\ Predictions}{\#Total\ Predictions} \quad (23)$$

where the number of correct predictions is the number of times the model correctly predicts the label of a token, and the total predictions refer to the total number of predictions made by the model.

Accuracy@k is an evaluation metric used in natural language processing to measure the accuracy of a model's top k predictions for a given task. In Vietnamese NLP, this metric is often used for tasks such as word segmentation, part-of-speech tagging, and named entity recognition.

$$Accuracy@k = \frac{1}{n} \sum_{i=1}^n \frac{1}{k} \sum_{j=1}^k \delta(y_{ij}, \hat{y}_{ij}) \quad (24)$$

where n is the number of instances, k is the number of predictions to consider, y_{ij} is the true label for the j th prediction of the i th instance, $\hat{y}_{ij}$ is the predicted label for the j th prediction of the i th instance, and δ is the Kronecker delta function which returns 1 if the predicted label matches the true label and 0 otherwise.

BLEU. The BLEU (bilingual evaluation understudy) (Papineni et al., 2002) is a widely used metric for evaluating the quality of machine translation output.

$$BLEU = \exp(\min(0, 1 - \frac{\sum_{n=1}^N \sum_{i=1}^m \min(c_i, \hat{c}_i^{(n)})}{\sum_{n=1}^N \sum_{i=1}^m \hat{c}_i^{(n)}})) \quad (25)$$

where N is the total number of sentences in the reference and hypothesis texts, m is the length of the longest sentence in the reference and hypothesis texts, c_i is the count of the i th n-gram in the reference text, and $\hat{c}_i^{(n)}$ is the count of the i th n-gram in the hypothesis text.

Exact Match. Exact Match, also known as EM score, is an evaluation metric used in natural language processing and machine learning to evaluate the performance of models on tasks such as question answering and machine reading comprehension.

$$EM = \frac{1}{N} \sum_{i=1}^N \delta(y_i, \hat{y}_i) \quad (26)$$

where N is the total number of examples, y_i is the ground truth label for the i th example, $\hat{y}_i$ is the predicted label for the i th example, and $\delta(y_i, \hat{y}_i)$ is the Kronecker delta function that returns 1 if $y_i = \hat{y}_i$, and 0 otherwise. The formula computes the proportion of the total number of examples in which the predicted label exactly matches the ground truth label.

ROUGE. ROUGE (Grusky, 2023) is a set of evaluation metrics commonly used for automatic text summarization. ROUGE-N measures the overlap between n-grams (sequences of n words) in the generated summary and the reference summary. ROUGE-L measures the longest common subsequence between the generated summary and the reference summary. ROGUE-S and ROGUE-W measure the overlap between the n-grams (contiguous sequences of words) in the generated summary and the reference summary. ROGUE-S uses unigrams (single words), while ROGUE-W uses longer n-grams up to a maximum length of N . The overall ROUGE score is the average of the ROUGE-N scores for different values of n . The ROUGE-N score is defined as:

Let c be the set of n -grams in the candidate summary, r be the set of n -grams in the reference summary, and m be the set of n -grams that appear in both c and r . Then, ROGUE-N is calculated as:

$$ROUGE-N = \frac{\sum_{g \in m} \min(\text{count}_c(g), \text{count}_r(g))}{\sum_{g \in c} \text{count}_c(g)} \quad (27)$$

where $\text{count}_c(g)$ denotes the number of times n -gram g appears in the candidate summary, and $\text{count}_r(g)$ denotes the number of times n -gram g appears in the reference summary.

The ROUGE-L score is defined as:

$$ROUGE-L = \frac{\sum_{s \in S} \text{lcs}(s, g(s))}{\sum_{s \in S} \text{len}(s)} \quad (28)$$

where $\text{lcs}(s, g(s))$ is the length of the longest common subsequence between the generated summary $g(s)$ and the reference summary s , and $\text{len}(s)$ is the length of the reference summary.

$$ROUGE-S = \frac{\sum_{(a,b) \in A} \text{LCS}(a,b)}{\sum_{(a,b) \in A} |b|} \quad (29)$$

where A is the set of sentence pairs, a is the reference sentence, b is the generated sentence, and $\text{LCS}(a,b)$ is the length of the longest common subsequence of a and b .

$$ROUGE-W = \frac{\sum_{i=1}^n W_i}{\sum_{i=1}^n c_i} \quad (30)$$

where n is the number of words, W_i is the weight assigned to word i , and c_i is the number of occurrences of word i in the reference summaries.

METEOR. Aside from BLEU, METEOR (Banerjee & Lavie, 2005) is another widely used metric for the machine-translation problem. The metric is based on the harmonic mean of unigram precision and recall, with

recall weighted higher than precision. It consists of two major components: a flexible monolingual word aligner with unique features such as stem match, synonym match, and paraphrase match, along with the exact match, and a scorer. Vietnamese is not yet supported for special alignment features other than exact match. As a result, the metric is limited to the Vietnamese language.

In the calculation of METEOR, the hypothesis sentences are first aligned to reference sentences. Alignments are then scored to produce sentence and corpus level scores following the formulas defined as:

$$F_{mean} = \frac{P \cdot R}{\alpha \cdot P + (1 - \alpha) \cdot R} \quad (31)$$

$$Penalty = \gamma \cdot \left(\frac{ch}{m} \right)^\beta \quad (32)$$

$$METEOR = F_{mean} \cdot (1 - Penalty) \quad (33)$$

where P and R denote uni-gram Precision and Recall respectively and are combined in the calculation of harmonic mean F_{mean} . The penalty of the given alignment is calculated using the counts of chunks ch and the matches m . The free parameters α , β , and γ are used to control the overall setting of the metric for calculating the final score.

CIDEr. CIDEr (Vedantam et al., 2015) is a metric proposed for evaluating image captioning tasks that has been shown to capture human judgment of consensus better than other known metrics (BLEU, ROUGE, METEOR). The CIDEr score for n-grams of length n is calculated using the average cosine similarity between the candidate phrase c_i and the reference sentences $S_i = \{s_{i1}, \dots, s_{im}\}$, which takes precision and recall into account, and is defined as:

$$g_k(s_{ij}) = \frac{h_k(s_{ij})}{\sum_{w_j \in \Omega} h_j(s_{ij})} \log \left(\frac{|I|}{\sum_{I_p \in I} \min(1, \sum_q h_k(s_{pq}))} \right) \quad (34)$$

$$CIDEr(c_i, S_i) = \sum_{n=1}^N \frac{w_n}{m} \sum_j \frac{g^n(c_i) \cdot g^n(s_{ij})}{\|g^n(c_i)\| \|g^n(s_{ij})\|} \quad (35)$$

where $g_k(s_{ij})$ is the TF-IDF weighting for each n-gram. $h_k(s_{ij})$ is the number of times an n-gram occurs in a reference sentence s_{ij} . $g^n(c_i)$ is a vector formed by $g_k(c_i)$ corresponding to all n-grams of length n and $\|g^n(c_i)\|$ is the magnitude of the vector $g^n(c_i)$. Similarly for $g^n(s_{ij})$.

SPICE. SPICE (Anderson et al., 2016) is another metric for automatic image caption evaluation and measuring the quality of image captions generated by models. The metric is defined over semantic graphs, which are used in many studies within the context of image and video retrieval systems to improve performance on complex queries. Given a candidate caption c and a set of reference captions $S = \{s_1, \dots, s_m\}$ associated with an image, the metric first performs semantic parsing to convert the candidate caption to a scene graph defined as:

$$G(c) = \langle O(c), E(c), K(c) \rangle \quad (36)$$

where $O(c)$ is the set of object mentions in c , $E(c)$ is the set of hyperedges representing relations between objects and $K(c)$ is the set of attribute types. The calculation of SPICE is then based on F-score formula, which is defined as:

$$P(c, S) = \frac{|T(G(c)) \otimes T(G(S))|}{T(G(c))} \quad (37)$$

$$R(c, S) = \frac{|T(G(c)) \otimes T(G(S))|}{T(G(S))} \quad (38)$$

$$SPICE(c, S) = \frac{1}{\frac{1}{P(c, S)} + \frac{1}{R(c, S)}} \quad (39)$$

where $T(G(c)) \triangleq O(c) \cup E(c) \cup K(c)$ return the logical tuples from the scene graph, similar for $T(G(S))$. Being an F-score, SPICE is simple to comprehend and interpret because it is naturally bounded between 0 and 1.

BERTScore. BERTScore (Zhang et al., 2020) is an automatic evaluation metric used to assess the quality of text generation systems. Similar to common metrics, BERTScore computes a similarity score for each token

in the candidate sentence with each token in the reference sentence. The difference is that it uses contextual embeddings, which allows it to better correlate with human assessments and outperform existing metrics in terms of model selection performance. Given a reference sentence $x = \{x_1, \dots, x_k\}$ and a candidate sentence $\hat{x} = \{\hat{x}_1, \dots, \hat{x}_l\}$, the metric uses BERT to represent the tokens and compute matching using cosine similarity. The recall, precision, and F1 scores are defined as:

$$P_{BERT} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} x_i^T \hat{x}_j \quad (40)$$

$$R_{BERT} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} x_i^T \hat{x}_j \quad (41)$$

$$BERTScore = \frac{1}{\frac{1}{P_{BERT}} + \frac{1}{R_{BERT}}} \quad (42)$$

GPTScore. GPTScore (Fu et al., 2024) is a training-free evaluation framework that assesses natural language generation quality using conditional probability from pre-trained language models. The core formula is:

$$GPTScore(h|S, d, a) = P(h|S, d, a) = \prod_{i=1}^m P(h_i|h_{1:i-1}, S, d, a) \quad (43)$$

where h is the hypothesis text, S is source context, d is task description, and a is the evaluation aspect. Length normalization using log-probabilities prevents bias toward shorter texts:[3]

$$GPTScore_{normalized} = \frac{1}{m} \sum_{i=1}^m \log P(h_i|h_{1:i-1}, S, d, a) \quad (44)$$

Unlike BLEU, ROUGE, or trained metrics like COMET, GPTScore enables flexible customizable evaluation across 22 aspects and 37 datasets spanning summarization, machine translation, dialogue, and data-to-text generation without requiring annotated training data. The framework supports source-to-hypothesis $P(\text{hypo}|\text{src})$ and reference-to-hypothesis $P(\text{hypo}|\text{ref})$ scoring, demonstrated across 19 models (80M-175B parameters) with strong correlation to human judgment. GPTScore outperforms traditional metrics by capturing semantic meaning, avoids the data annotation bottleneck of learned metrics, and provides greater interpretability than chain-of-thought alternatives like G-EVAL. Despite limitations including LLM dependency and performance variation by model and aspect definition quality, GPTScore has become foundational for scalable NLG evaluation in localization, summarization, dialogue assessment, and research benchmarking.

4. Challenges and research directions

The challenges facing Vietnamese NLP form a mutually reinforcing constraint system: linguistic complexity drives data scarcity, which limits model capability, which in turn restricts evaluation standardization and real-world deployment. For each challenge identified below, we directly specify actionable research directions, providing both a diagnostic and a prescriptive roadmap for the community.

4.1. Linguistic characteristics and dialectal constraints

Vietnamese possesses unique typological features that pose distinct challenges for computational modeling, ranging from morpho-syntactic isolation to deep semantic and regional complexity:

- **Typological Isolation and Segmentation Ambiguity:** As an analytic language lacking inflectional morphology, Vietnamese relies on word order rather than grammatical markers. Its monosyllabic orthography where whitespace delimits syllables rather than semantic words results in significant word segmentation ambiguity. This lack of explicit boundaries complicates foundational NLP tasks like tokenization and syntactic parsing, as errors here propagate downstream. Vietnamese is also a heavily pro-drop language, frequently

omitting subjects or objects when pragmatically inferable, creating complex topic-comment structures that resist standard dependency parsing paradigms designed for rigid SVO languages.

- **Phonemic Tonal System:** The standard dialect employs six phonemic tones ngang (level), huyền (falling), sắc (rising), hỏi (dipping), ngã (broken), nặng (heavy) where pitch directly determines lexical identity: ma (ghost), má (mother), mà (but), mã (tomb), mạ (rice seedling), and mã (horse) differ only in tonal marker.
- **Underexplored Semantics and Parsing Complexity:** There is a critical scarcity of research into Vietnamese semantic parsing. As a high-context language featuring frequent argument omission (pro-drop) and complex topic-comment structures, Vietnamese is difficult to map to standard logical formalisms. The absence of annotated corpora for tasks like Abstract Meaning Representation (AMR) limits current models to surface-level pattern matching rather than deep reasoning.
- **Dialectal Heterogeneity and Analytical Absence:** Despite profound regional heterogeneity (Northern, Central, Southern), the current landscape is marked by a conspicuous absence of dialect analysis. Existing resources disproportionately favour the standard Northern dialect, creating a representational gap with two critical implications:
 - (a) *Pedagogical Limitation:* Automated systems present a homogenised model of Vietnamese, failing to equip learners or algorithms with the competence to understand diverse regional speakers.
 - (b) *Preservation Risk:* The lack of systematic documentation for regional variations threatens local cultural identities, as unique dialectal features face attrition without digital archival.

Research Direction. Future work should construct AMR-annotated corpora for Vietnamese to provide richer structural supervision beyond surface-level dependency trees. Joint segmentation-tagging models should replace sequential pipelines to eliminate cascade errors. For the tonal system, diacritic restoration must be integrated as a mandatory pre-processing step in all general-purpose pipelines, and models such as ViSoBERT (Nguyen et al., 2023e) whose tokenizer is designed natively for informal Vietnamese “teencode” should be adopted for all social media NLP tasks. Dialectal normalisation frameworks that map non-standard regional inputs into standardised Vietnamese representations prior to LLM inference represent the practical path toward equitable AI access without requiring per-dialect model retraining.

4.2. Low resource availability

The scarcity of high-quality annotated data and corpora constitutes a primary obstacle for Vietnamese NLP. This deficiency, stemming from multifaceted factors including linguistic complexity, insufficient funding, and a limited research workforce, necessitates resource-intensive manual annotation and is further compounded by a lack of tool standardisation. To mitigate these constraints, researchers are increasingly leveraging approaches like transfer learning, domain adaptation, and unsupervised learning. While these strategies have yielded promising results, sustained research and development remain imperative to fully surmount the resource deficit and facilitate the widespread deployment of Vietnamese NLP systems.

Research Direction. Cross-lingual transfer from English remains the practical recommendation for bootstrapping Vietnamese NLP systems in under-resourced task categories. LLM-based synthetic data generation provides the most scalable corpus construction path given current annotation costs. Multi-institutional collaborations following the VLSP shared-task model should be formalised as standing annual initiatives to ensure sustained corpus growth across all task categories identified in this survey.

Table 22
Overview of Vietnamese NLP toolkits and supported tasks.

Toolkit	Tasks	Architecture	License
RDRSegmenter	WS, POS	Rule-list (RDR)	Open-source
VnCoreNLP	WS, POS, NER, Parsing	BiLSTM-CRF	Open-source
pyvi	Word Segmentation	Neural	Open-source
PhoNLP	POS, NER, Parsing	PhoBERT multi-task	Apache 2.0
Trankit	WS, POS, NER, Parsing	XLM-R	Apache 2.0

4.3. Lack of necessary tools

A significant gap exists between resource-rich languages like English and Vietnamese regarding available NLP toolkits. Vietnamese relies primarily on standard tools like VnCoreNLP (Vu et al., 2018) and Underthesea,⁴ which achieve decent accuracy for core tasks. To improve performance in specific contexts, recent research has introduced targeted tokenisation models: CafeBERT (Do et al., 2024b) handles open-domain data, while ViSoBERT (Nguyen et al., 2023e) is optimised to handle the irregularities found in social media text. Table 22 provides a complete overview of available Vietnamese NLP toolkits and their supported tasks.

Research Direction. No existing toolkit yet provides an end-to-end pipeline spanning diacritic restoration, word segmentation, core NLP tasks, and generative capabilities. The unification of VnCoreNLP-level core processing, ViSoBERT-level informal text handling, and BARTpho-level generation into a single modular toolkit represents an urgent and high-impact software engineering target. Standardised Python packaging (following the spaCy pipeline model) would substantially lower the barrier for downstream developers and international researchers working with Vietnamese.

4.4. Data scarcity and benchmarking deficiencies

Compared to high-resource languages like English, which benefit from massive datasets such as SNAP (80 million reviews) (Leskovec et al., 2016) or SQuAD 2.0 (150,000+ QA pairs) (Rajpurkar et al., 2018), Vietnamese resources remain significantly limited in scale. This scarcity poses a major research bottleneck, increasingly necessitating the use of synthetic data generation to augment training corpora and improve model generalisation. The reliance on limited real data and synthetic augmentation highlights a critical gap: the absence of a standardised comprehensive benchmark suite. Unlike the GLUE (Wang et al., 2018) or SuperGLUE (Wang et al., 2019) benchmarks in English, Vietnamese lacks a unified evaluation framework designed to assess the full spectrum of linguistic abilities. Without such a benchmark, it remains challenging to rigorously evaluate whether models have achieved true language understanding or are merely overfitting to small, fragmented datasets.

Research Direction. We recommend two parallel investments. First, the community should extend VLUE (Do et al., 2024b) into a unified Vietnamese GLUE-style benchmark covering the full NLU and NLG task spectrum with a standardised, versioned public leaderboard. Second, the community should construct a 100,000+ sample Vietnamese instruction-tuning dataset spanning legal, medical, and educational domains, enabling supervised fine-tuning of open-weight models via Parameter-Efficient Fine-Tuning for secure, localised enterprise deployment.

4.5. Scarcity of knowledge-intensive and reasoning datasets

The limited availability of Vietnamese datasets stems from multifaceted constraints, including insufficient funding, the absence of standardised protocols for dataset creation, and the inherent complexity

of the language itself. Developing high-quality resources is a resource-intensive process requiring rigorous manual labour from data collection to fine-grained annotation which creates a significant bottleneck for the field.

Critically, this scarcity is not merely quantitative but qualitative. Current benchmarks are predominantly characterised by “shallow” complexity, focusing on surface-level tasks such as standard sentiment analysis or Named Entity Recognition (NER), while lacking datasets that require deep semantic understanding such as Natural Language Inference (NLI) (Huynh et al., 2022) or commonsense reasoning. This deficiency forces models to rely on superficial statistical artefacts rather than logical deduction, leading to significant brittleness and adversarial vulnerability.

A further dimension of this problem is **evaluation inconsistency**: many Vietnamese NLP papers define their own random train/dev/test splits rather than using canonical splits, and results are routinely reported without specifying which word segmenter (RDRSegmenter, VnCoreNLP, or pyvi) was used for preprocessing. Since different segmenters produce non-equivalent tokenizations, reported scores are frequently non-comparable even across papers using the same dataset and model architecture. The proposed VLUE extension directly addresses this by mandating standardised splits and a fixed preprocessing pipeline as conditions of leaderboard submission.

Research Direction. Concerted efforts are required to elevate dataset complexity through multi-institutional collaborations to construct reasoning-oriented and adversarial benchmarks. This includes extending ViFactCheck (Hoa et al., 2025) to expose brittleness in state-of-the-art systems and developing multimodal misinformation benchmarks combining image and text claims. Establishing rigorous standardisation bodies for annotation processes is essential to ensure future datasets possess the consistency and depth necessary for robust model development.

4.6. Delayed adoption of large language models (LLMs)

The adoption of LLMs in Vietnamese NLP trails significantly behind high-resource languages, primarily due to the scarcity of large-scale corpora and computational constraints. While open-source efforts such as PhoGPT (Nguyen et al., 2023d), Vistral,⁵ and Sailor represent meaningful milestones (see §3.4), they lack the maturity of English or Chinese counterparts. Critically, the field suffers from a severe absence of domain-specific models for vertical sectors like law, medicine, or finance. This gap is driven by the proprietary and sensitive nature of domain data, which complicates curation. Furthermore, generic Vietnamese models frequently exhibit hallucinations when applied to specialised contexts due to a lack of domain-specific instruction-tuning datasets. Consequently, the deployment of reliable downstream applications in high-stakes fields remains stagnant, highlighting an urgent need for research into parameter-efficient fine-tuning and domain-adaptive pre-training strategies.

Research Direction. We recommend developing Small Language Models (SLMs, ≤4B parameters) via a two-stage training pipeline: (1) domain-adaptive continual pre-training on high-quality domain text, followed by (2) PEFT using LoRA on task-specific supervised data. Indigenous multimodal integration must also be advanced: future datasets should capture native Vietnamese visual environments street scenes, government forms, invoices supporting Layout-Aware architectures such as LiGT (Le et al., 2025) for Vietnamese document intelligence. These domain-specific SLMs, deployed on-premise, address the data sovereignty requirements of Vietnamese legal and medical organisations that prohibit transmission of sensitive data to external API endpoints.

⁴ <https://github.com/undertheseanlp/underthesea>

⁵ <https://github.com/VinAIRResearch/Vistral>

Table 23
Vietnamese NLP challenges, impacted tasks, and research directions.

Challenge	Primary Impacted Tasks	Research Direction
Typological isolation; word segmentation ambiguity	POS, NER, Parsing, MT	Joint segmentation-tagging; syllable-level BPE; AMR corpus construction
Six-tone tonal/diacritic system	OCR, ASR, Social NLP, SA	Diacritic restoration pipeline; ViSoBERT for informal text
Dialectal heterogeneity (N/C/S)	ASR, Dialogue, Sentiment, MT	Dialect-labelled corpora; normalisation pipeline as pre-processing layer
Low resource availability	All tasks	Cross-lingual transfer; LLM synthetic data; annual VLSP campaigns
Lack of NLP toolkits	All pipeline tasks	Unified end-to-end toolkit; standardised Python packaging
Benchmark fragmentation	Model comparison, reproducibility	Unified GLUE-style leaderboard (VLU extension)
Shallow/reasoning dataset scarcity	NLI, MRC, Commonsense, Fact-checking	Adversarial benchmarks; ViFactCheck extension
LLM instruction-tuning lag	Generative QA, Legal, Medical NLP	PEFT (LoRA) on domain SLMs ($\leq 4B$ params); 100K+ instruction dataset
Hallucination in generation	Legal, Medical, Fact-checking	RAG grounding; LiGT for document intelligence
Explainability gap	Legal, Medical, Financial NLP	Rationale annotation; source-attributed RAG; XAI interface tools

4.7. Explainability and interpretability

According to [Arrieta et al. \(2020\)](#), interpretability is the ability to provide the decision-making process of the system to humans in understandable terms, while explainability is the ability to provide a decision-making process that is not only accurate for the computer but also comprehensible for humans. Explainability and Interpretability are important for constructing intelligent systems in high-stakes domains such as legal, healthcare, and finance because users want to understand how the system returns a certain output. Acquiring explainability for AI systems improves user trustworthiness and is a prerequisite for regulatory compliance in the Vietnamese legal framework.

Research Direction. We recommend a three-component approach: (i) construct rigorous **adversarial evaluation datasets** by extending ViFactCheck ([Hoa et al., 2025](#)) to expose brittleness in current systems; (ii) integrate **Retrieval-Augmented Generation (RAG) frameworks** that ground generated outputs in verifiable Vietnamese source documents with complete answer traceability; and (iii) develop **XAI interfaces** providing human-interpretable rationale for model decisions a non-negotiable requirement for legal and medical NLP deployment where black-box generation is unacceptable.

5. Related work

Prior survey efforts in Vietnamese NLP have been valuable but largely task-specific or narrowly scoped. Early overviews focused primarily on foundational linguistic resources and word segmentation toolkits, while more recent efforts addressed individual application domains such as sentiment analysis or machine translation in isolation. To our knowledge, no prior work has simultaneously synthesized Vietnamese NLP datasets, the full architectural evolution across all paradigms (statistical through LLM-era), standardized benchmark performance, structured linguistic challenge analysis, and a concrete forward-looking research roadmap within a single reference. The present survey is distinguished by its breadth, analytical depth, and explicit treatment of the generative foundation model era a methodological transition insufficiently addressed in prior Vietnamese NLP literature.

Vietnamese survey

Vietnamese word segmentation ([Cong et al., 2016](#)) presents a comprehensive review of the current state-of-the-art methodologies and systems employed for word segmentation in the Vietnamese language. The objective of this study is to provide an encompassing overview of the various stages involved in the process, starting from the creation of relevant corpora to the development of specialized toolkits. Specifically, the paper delves into the initial stage of corpus construction, explores the different approaches utilized to address the task of word segmentation, and examines the existing toolkits available for effectively seg-

menting words within Vietnamese sentences. Through this systematic analysis, the paper aims to offer valuable insights into the advancements and challenges associated with word segmentation in Vietnamese, facilitating further research and development in this critical area of natural language processing.

Vietnamese Automatic Speech Recognition ([Nga et al., 2021](#)) focuses on the development of a Vietnamese speech recognition system that uses neural network models to improve accuracy and performance. The authors describe the various steps involved in building the system, including data collection, preprocessing, and model training.

Sentiment Analysis ([Phan et al., 2021](#)) proposes a deep learning approach for sentiment analysis in Vietnamese text. The proposed model utilizes a convolutional neural network (CNN) architecture to learn informative features from text data and a long short-term memory (LSTM) network to capture sequential information. The authors evaluate their model on several benchmark datasets and achieve competitive results compared to other state-of-the-art approaches.

Vietnamese method survey

Micro-text augmentation techniques survey ([Duong & Hoang, 2023](#)) introduces a text augmentation method for generating annotated training data automatically, eliminating the need for user intervention. The study focuses on exploring cross-language methods to enhance data and evaluates the benefits and drawbacks of each approach for Vietnamese language processing. The results demonstrate that text augmentation yields competitive performance and offers time and cost savings in data construction.

Multi-class classifiers survey ([Duong & Truong Hoang, 2019](#)) propose an approach to classify Vietnamese text data into multiple classes using a multi-class logistic regression model. The authors explore several feature extraction methods such as bag-of-words, n-gram, and character n-gram, along with various feature selection techniques including chi-square and mutual information.

6. Conclusions

This paper presents a comprehensive survey of Vietnamese natural language processing (NLP) techniques, datasets, and evaluation metrics in a systematic manner. The study addresses the primary challenges and research advancements in conceptualizing, planning, and solving problems in Vietnamese NLP. Additionally, we propose a unified taxonomy that categorizes Vietnamese NLP strategies into three approaches: Neural Word Representations models, Neural Language models, Transformer Language models and Multimodal models. We anticipate that this taxonomy will aid in categorizing future studies and provide a deeper understanding of methodologies while addressing the difficulties in Vietnamese NLP. The survey offers clear summaries and comparisons of various Vietnamese NLP methodologies, presenting a comprehensive and general view of current work and the current process of Vietnamese

NLP. Furthermore, we provide a foundation for structured benchmarking by assembling popular datasets and open-source applications to advance future studies in this field. The current trends and future prospects are highlighted based on the survey results, and we expect many applications from various domains to profit from Vietnamese NLP in the coming years.

CRedit authorship contribution statement

Khiem Vinh Tran: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft; **Triet Minh Thai:** Investigation, Writing – original draft; **Son T. Luu:** Investigation, Writing – original draft; **Kiet Van Nguyen:** Conceptualization, Data curation, Supervision, Writing – review & editing; **Ngan Luu Thuy Nguyen:** Validation, Supervision, Writing – review & editing.

Data availability

Data will be made available on reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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